Chapter 1
Computer Networks & the Internet

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Chapter 1: Computer Networks & the Internet

Our Goal:

- Chapter 1 presents a broad overview of computer networking and the Internet.
- To paint a broad picture and set the context for the rest of the book
- More depth, detail *later* in course

Overview:

- what's the Internet?
- what's a protocol?
- network edge; hosts, access net, physical media
- network core: packet/circuit switching, Internet structure
- performance: loss, delay, throughput
- protocol layers, service models
- security

Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

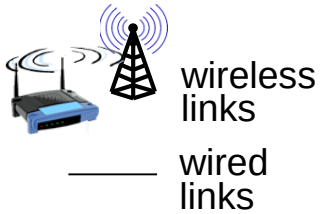
1.7 history

What is the Internet?

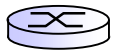
- Two ways to answer this question
  1. **The nuts and bolts of the Internet :**
     the basic hardware and software components that make up the Internet
  2. The Internet in terms of a **networking infrastructure** :
     a networking infrastructure that provides services to distributed applications.

What's the Internet: "nuts and bolts" view

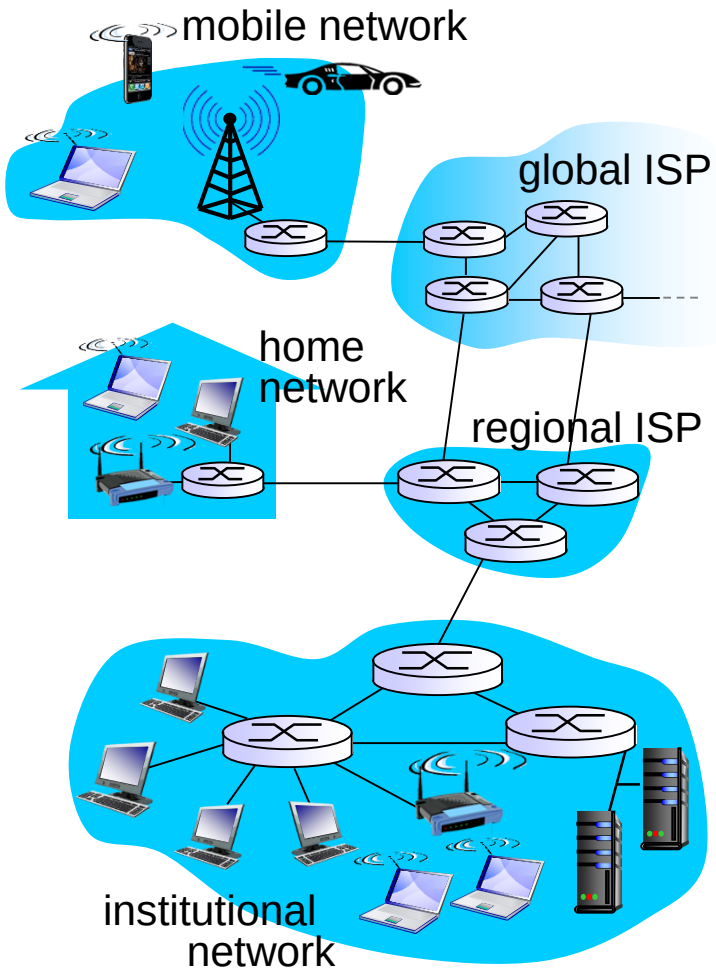
- millions of connected computing devices:
  - ***hosts*** = ***end systems***
  - running *network apps*



- *communication links*
  - fiber, copper, radio, satellite
  - transmission rate: *bandwidth*



- *Packet switches:* forward packets (chunks of data)
  - *routers* and *switches*



"Fun" internet appliances

IP picture frame
http://www.ceiva.com/

Web-enabled toaster +
weather forecaster

Tweet-a-watt:
monitor energy use

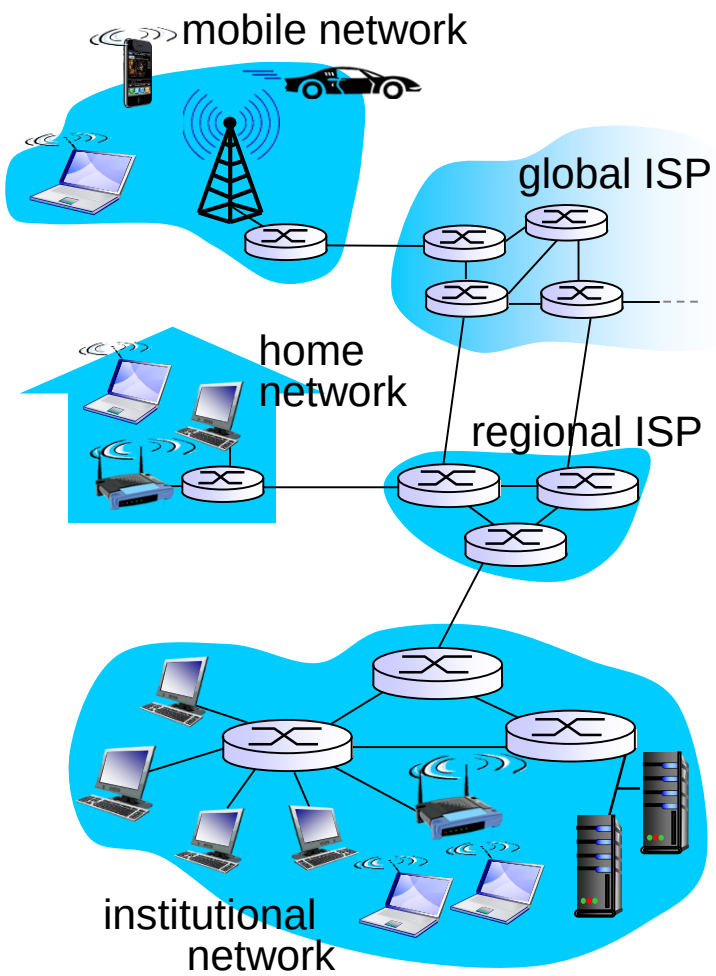
Internet
refrigerator

Slingbox: watch,
control cable TV remotely

Internet phones

What's the Internet: "nuts and bolts" view

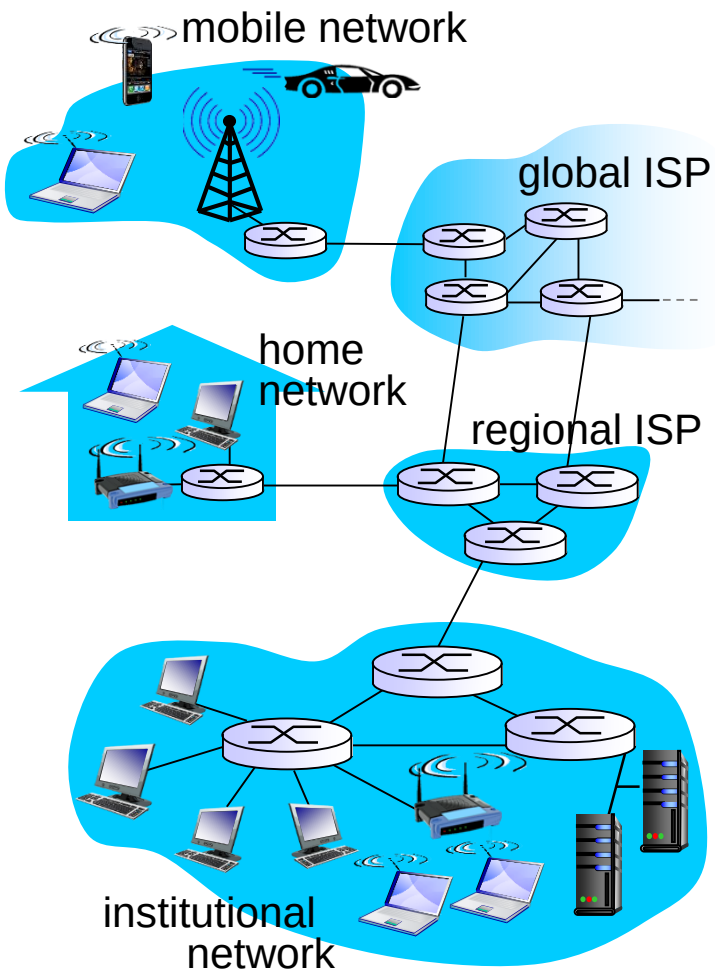
- *Internet:* "network of networks"
  - Interconnected ISPs (Internet Service Providers)
- *protocols* control sending, receiving of messages
  - e.g., TCP, IP, HTTP, Skype, 802.11
    - TCP: Transmission Control Protocol
    - IP: Internet Protocol
- *Internet standards*
  - RFC: Request for comments
  - IETF: Internet Engineering Task Force



What's the Internet: a service view

- *Infrastructure that provides services to applications:*
  - Web, VoIP, email, games, e-commerce, social nets, …
  - Called *distributed applications* due to multiple end systems that exchange data with each other.

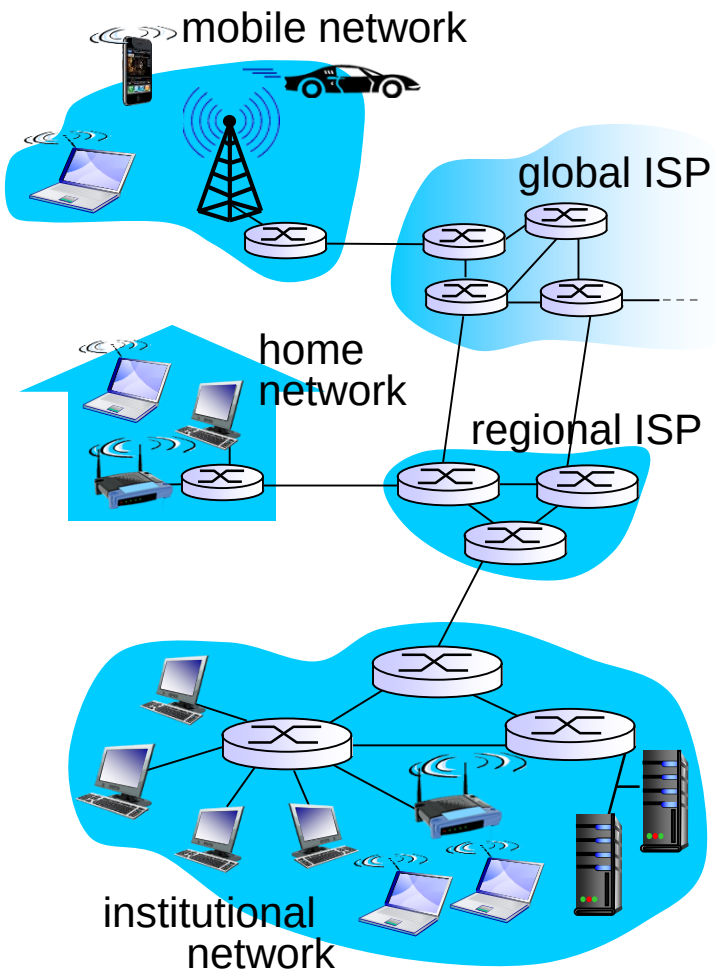
- *Provides programming interface to apps*
  - hooks that allow sending and receiving app programs to "connect" to Internet
  - provides service options, analogous to postal service



What's the Internet: a service view

- End systems attached to the Internet provide an *Application Programming Interface (API)* that specifies how a program running on one end system asks the Internet infrastructure to deliver data to a specific destination program running on another end system.

- This Internet API is a set of rules that the sending program must follow so that the Internet can deliver the data to the destination program.



What is a protocol?

human protocols:

- "what's the time?"
- "I have a question"
- introductions

… specific msgs sent

… specific actions taken when msgs received, or other events

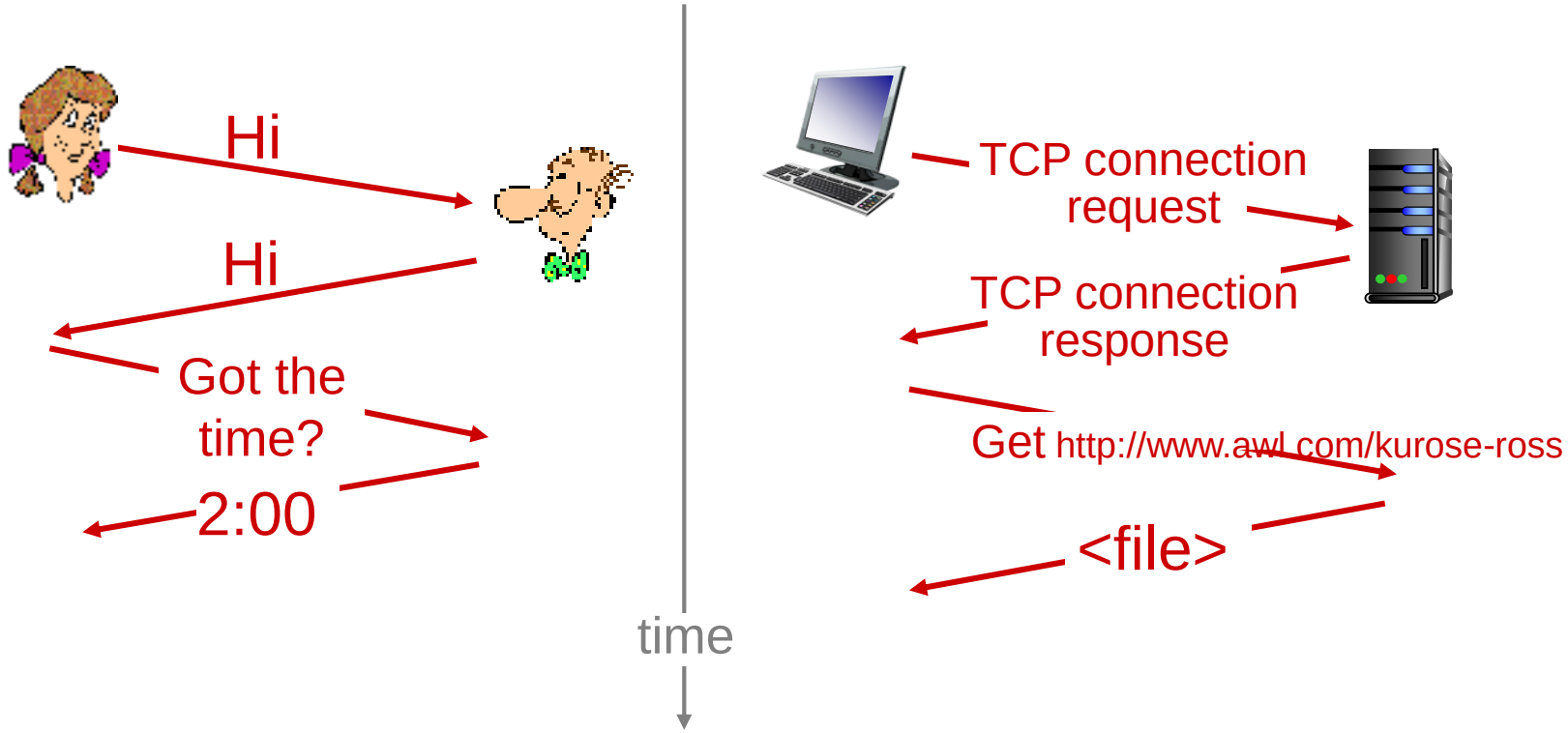
network protocols:

- machines rather than humans
- all communication activity in Internet governed by protocols

*A **protocol** defines the format, the order of msgs exchanged between two or more communicating entities, and the actions taken on msg transmission, receipt*

What is a protocol?

a human protocol and a computer network protocol:



Q: other human protocols?

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- end systems, access networks, links

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- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

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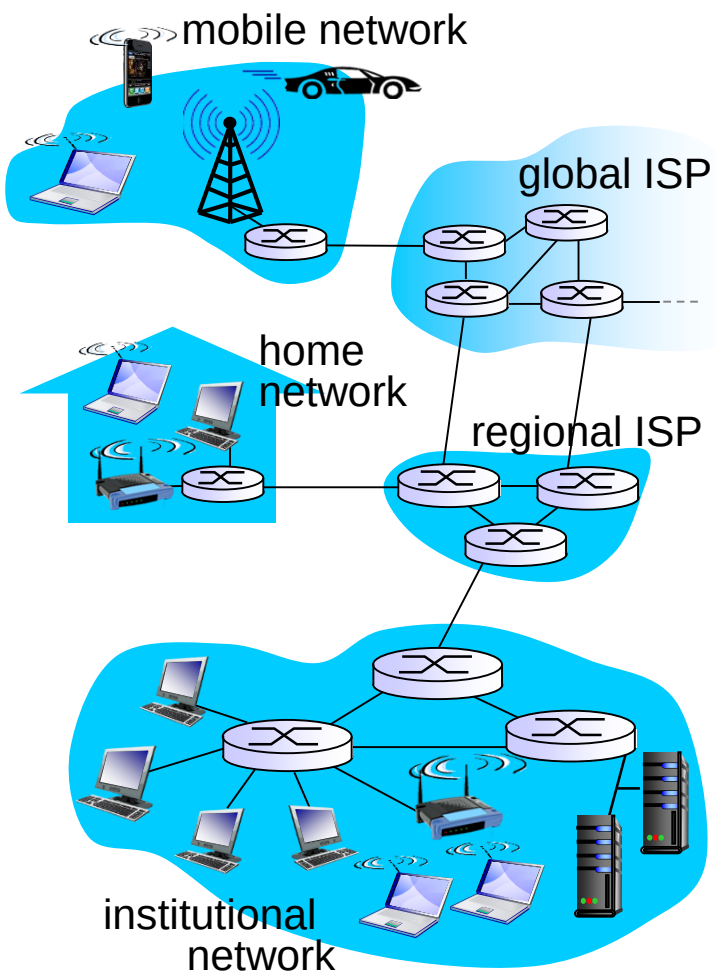
A closer look at network structure:

- *network edge:*
  - end system = host (because they host application programs)
  - hosts: **clients** and **servers**
  - servers often reside in data centers
- *access networks, physical media:* wired, wireless communication links
- *network core:*
  - interconnected routers
  - network of networks



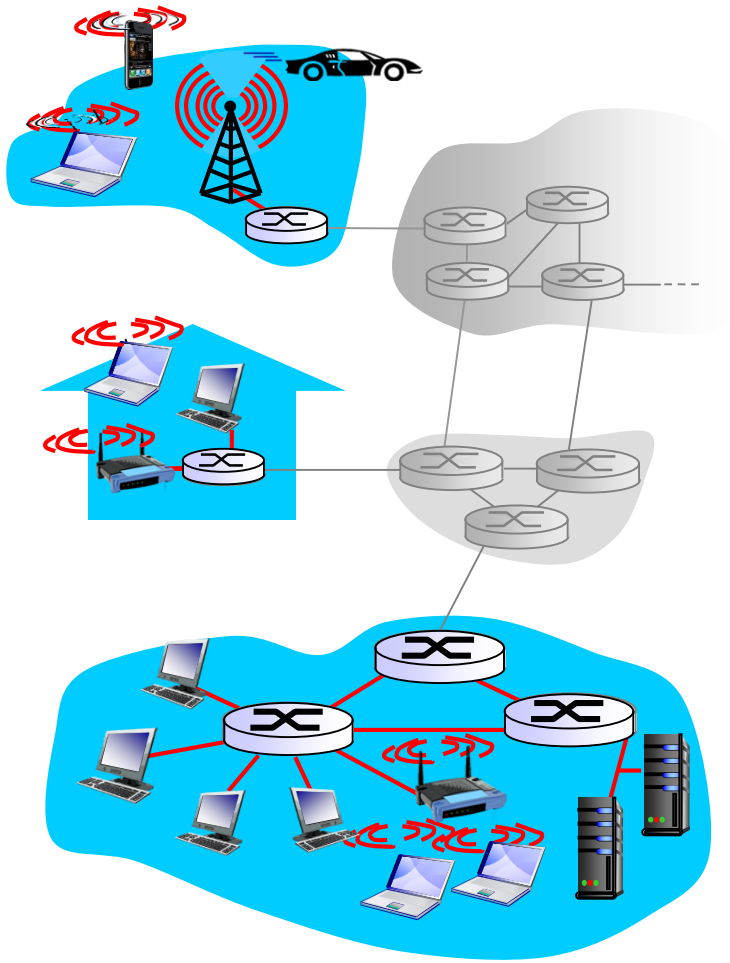
Access networks

The access network

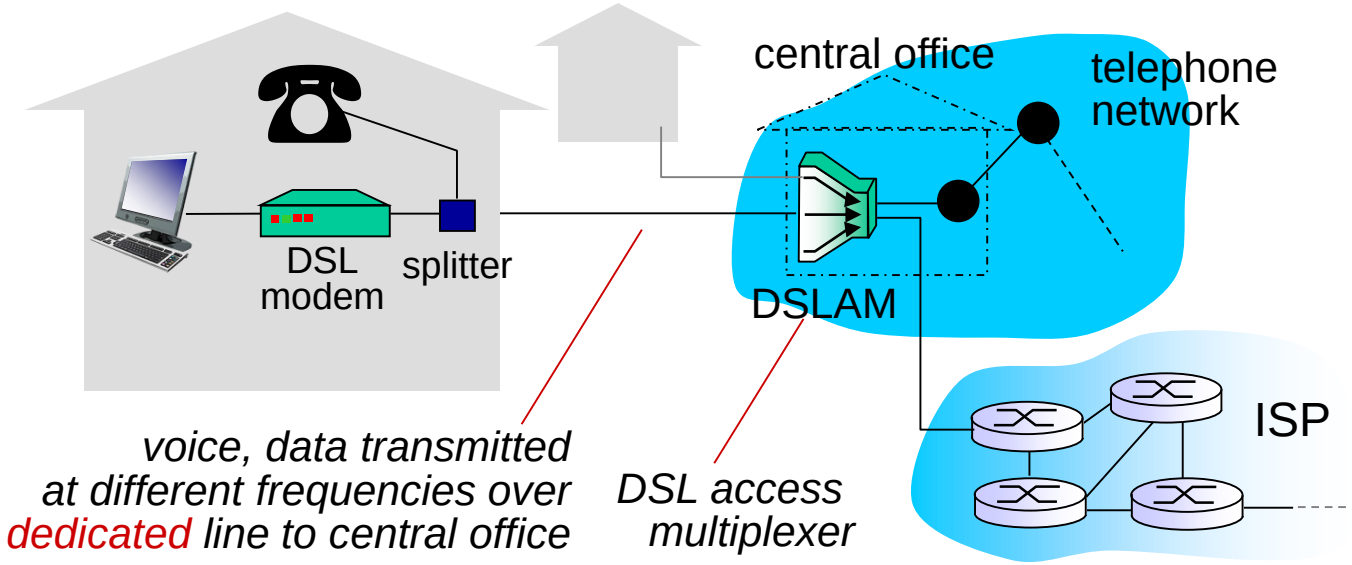
- the network that physically connects an end system to the first router (also known as the "edge router") on a path from the end system to any other distant end system.

keep in mind:

- bandwidth (bits per second) of access network?
- shared or dedicated?

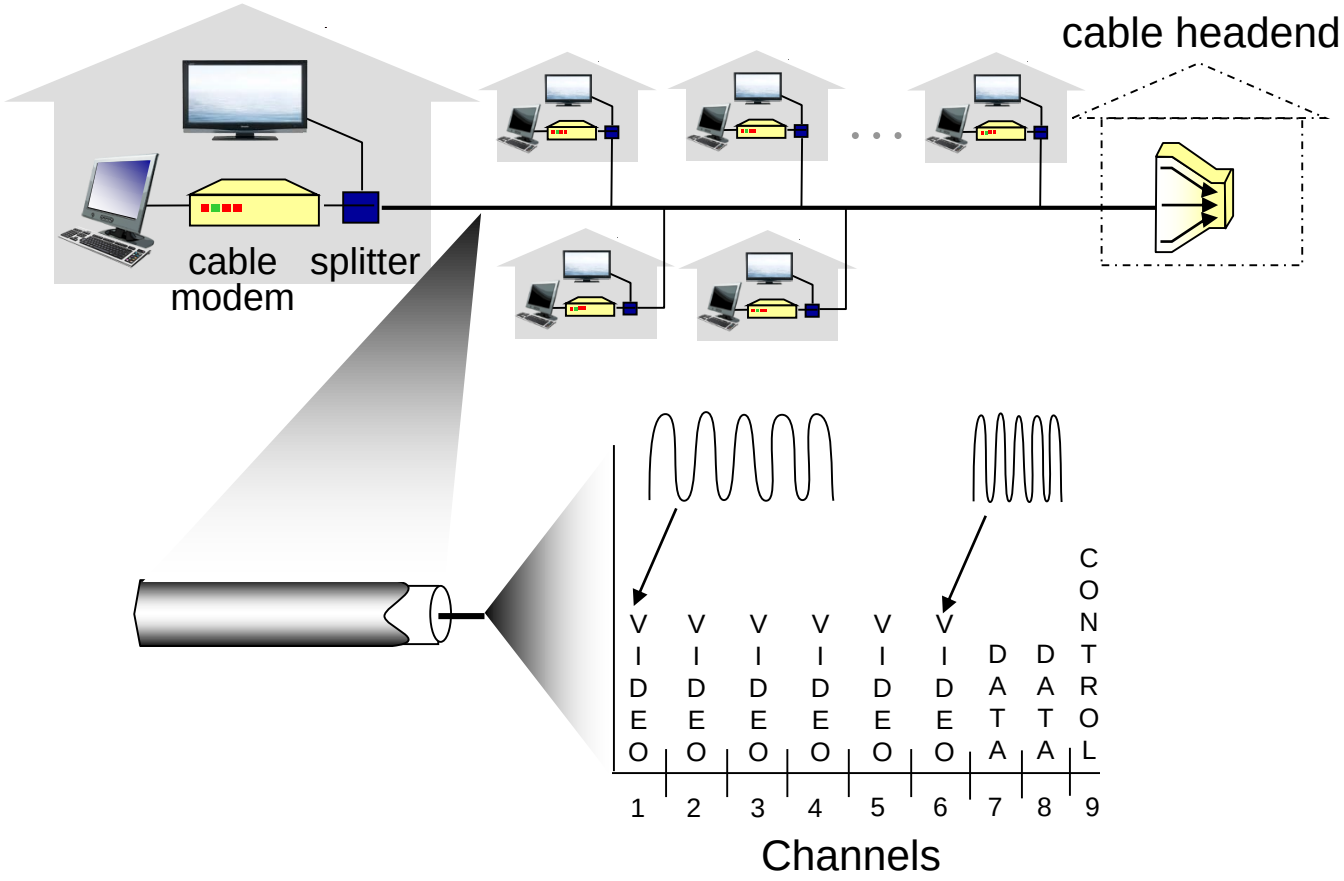


Home Access: digital subscriber line (DSL)



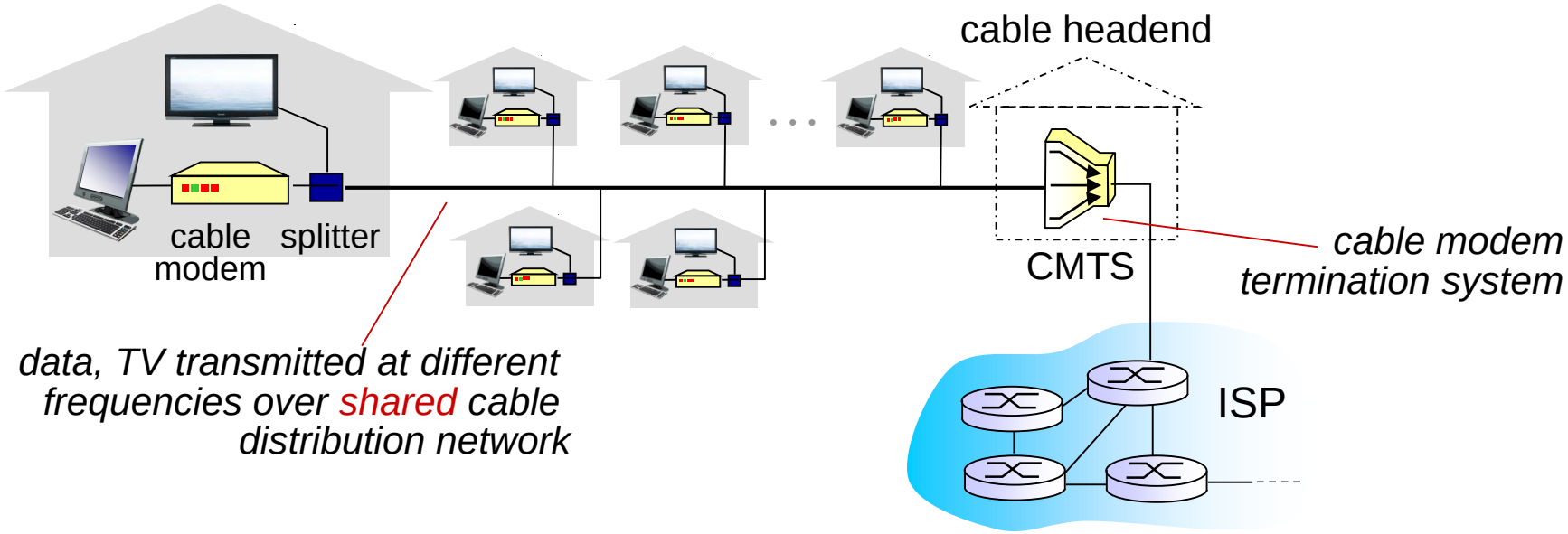
- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- < 24 Mbps downstream transmission rate (typically < 10 Mbps)

Home Access: cable network



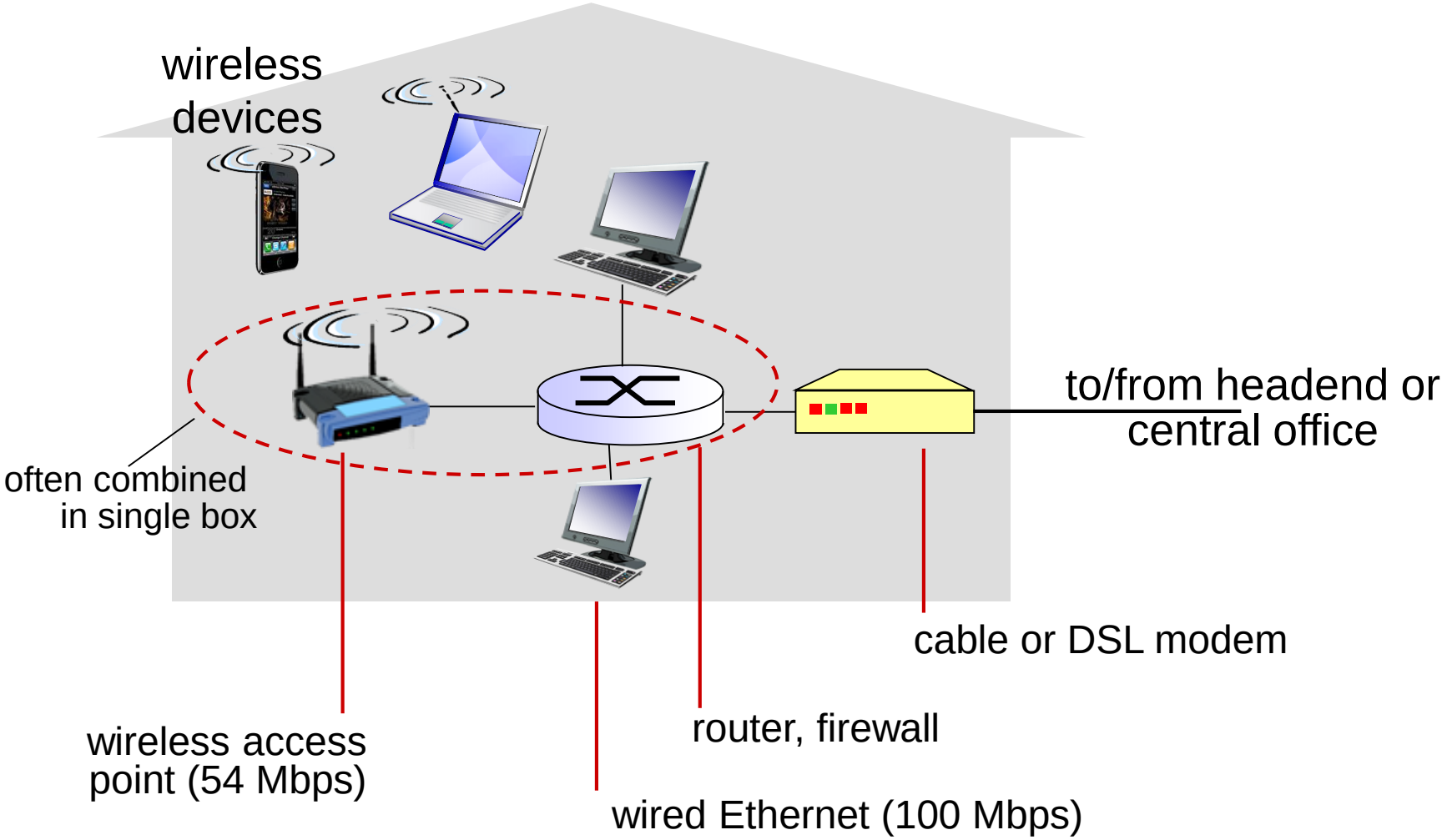
frequency division multiplexing: different channels transmitted in different frequency bands

Home Access: cable network

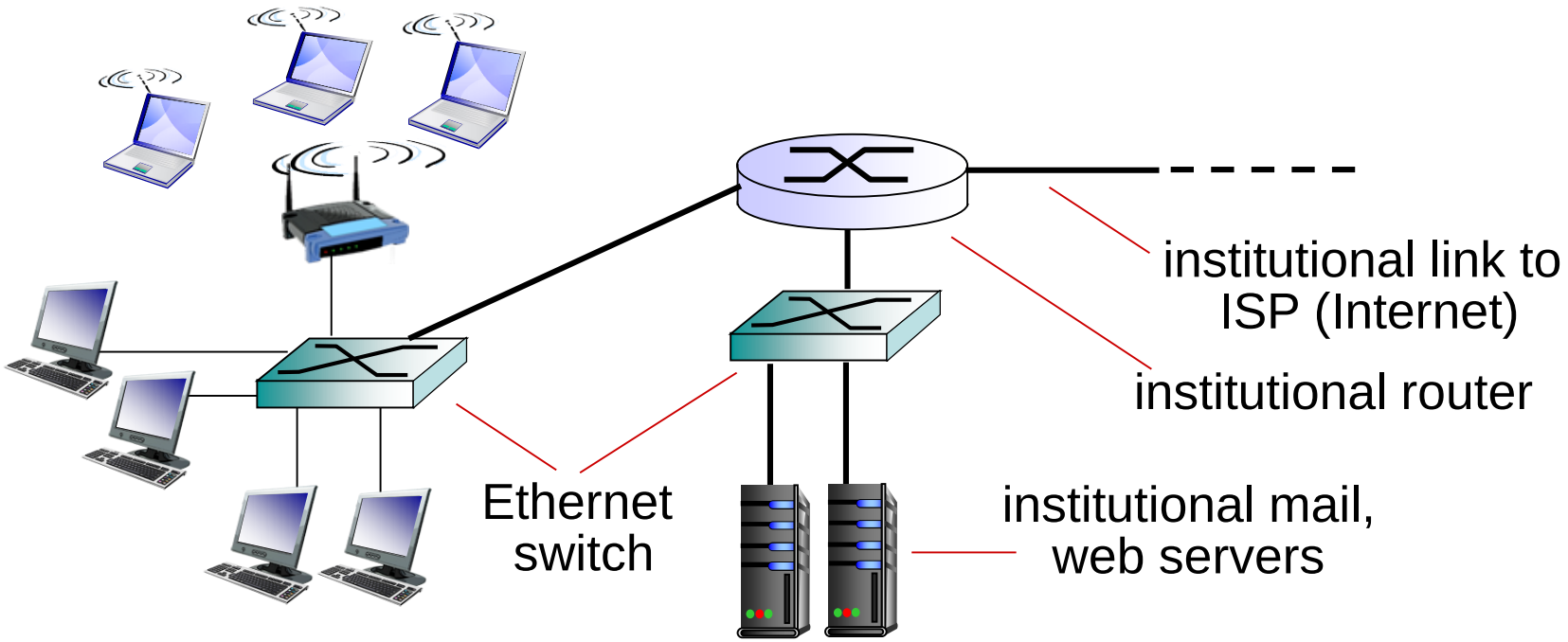


- HFC: hybrid fiber coax
  - Both fiber and coaxial cable are employed
  - Asymmetric: up to 30Mbps downstream transmission rate, 2 Mbps upstream transmission rate
- Network of cable, fiber attaches homes to ISP router
  - homes *share access network* to cable headend
  - unlike DSL, which has dedicated access to central office

Access in the home network



Enterprise access networks (Ethernet)



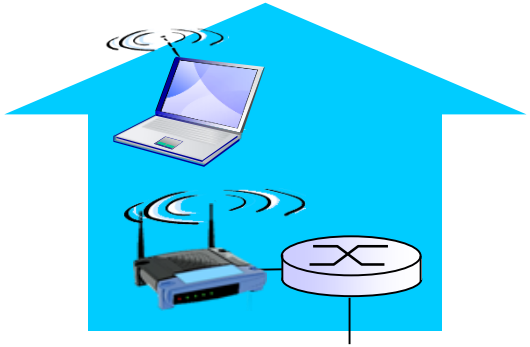
- typically used in companies, universities, etc
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- today, end systems typically connect into Ethernet switch

Wireless access networks

- shared *wireless* access network connects end system to router
  - via base station aka "access point"

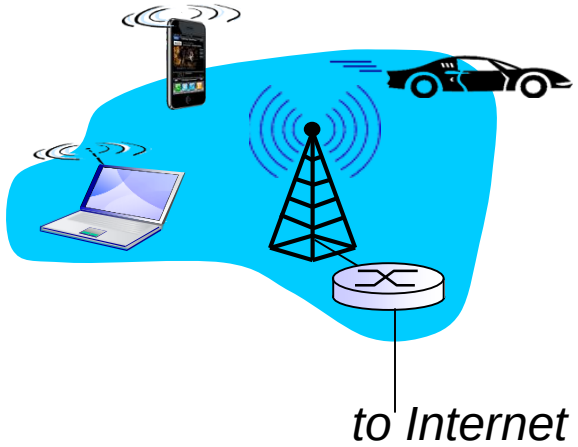
wireless LANs:

- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate



wide-area wireless access

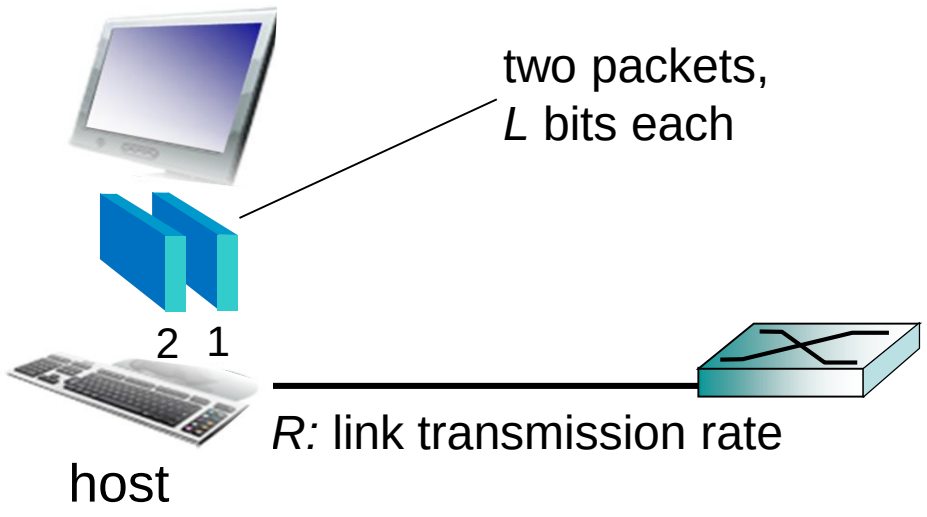
- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE



Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L\ (\text{bits})}{R\ (\text{bits/sec})}$$

Physical media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

Physical media

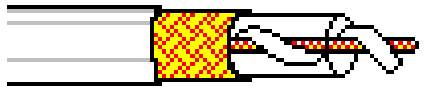
Twisted-Pair (TP) Copper Wire

- The least expensive and most commonly used guided transmission medium
- For over a hundred years it has been used by telephone networks
- More than 99% of the wired connections
- Twisted pair consists of two insulated copper wires, each about 1 mm thick, arranged in a regular spiral pattern
- **Unshielded twisted pair (UTP)** is commonly used for computer networks within a building, that is, for LANs.
  - Data rates for LANs using twisted pair today range from 10 Mbps to 10 Gbps

Physical media: coax, fiber

coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple channels on cable
  - HFC



fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (e.g., 10' s-100' s Gpbs transmission rate)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise

Physical media: radio

- signal carried in electromagnetic spectrum
- no physical "wire"
- bidirectional
- propagation environment effects:
  - reflection
  - obstruction by objects
  - interference

radio link types:

- terrestrial microwave
  - e.g. up to 45 Mbps channels
- LAN (e.g., WiFi)
  - 11Mbps, 54 Mbps
- wide-area (e.g., cellular)
  - 3G cellular: ~ few Mbps
- satellite
  - Kbps to 45Mbps channel (or multiple smaller channels)
  - 270 msec end-end delay
  - geosynchronous versus low altitude

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- packet switching, circuit switching, network structure

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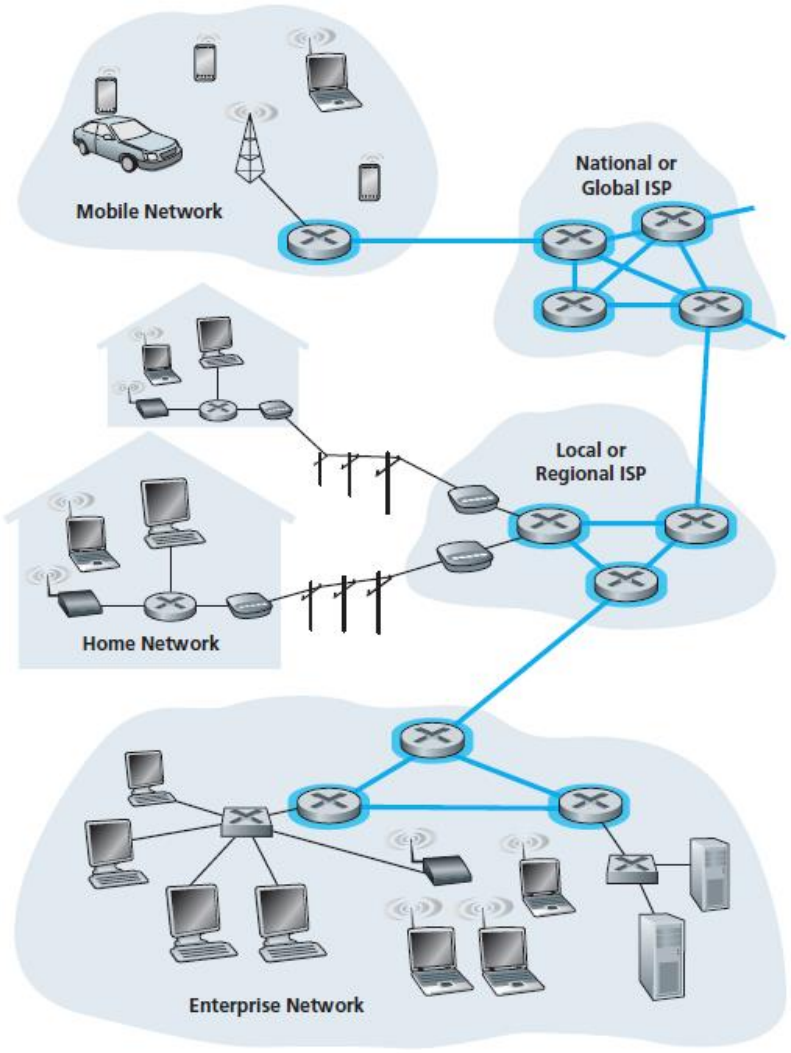
The network core

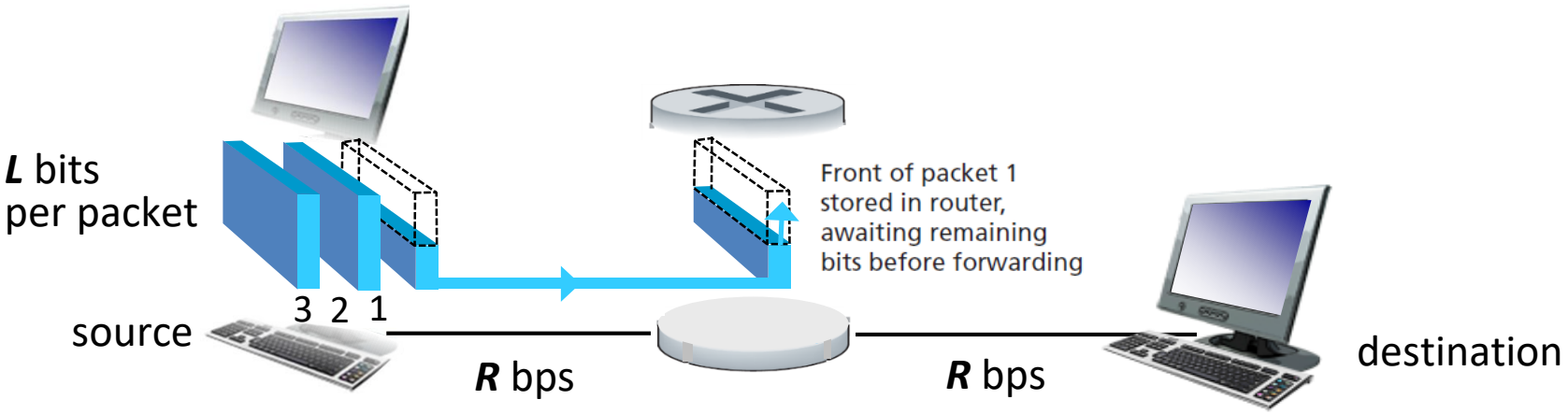
- Network core: <u>the mesh of packet switches and links</u> that interconnects the Internet's end systems



Packet-switching

- In a network application, end systems exchange **messages** with each other.
  - Messages can contain anything the application designer wants
- To send a message from a source end system to a destination end system, the source breaks long messages into smaller chunks of data known as **packets**.
- Between source and destination, each packet travels through communication links and **packet switches**
  - Two predominant types: **routers, link-layer switches**
- Packet-switching:
  - forward packets from one packet switch to the next, across links on path from source to destination
  - each packet transmitted at full link capacity

Packet-switching: store-and-forward



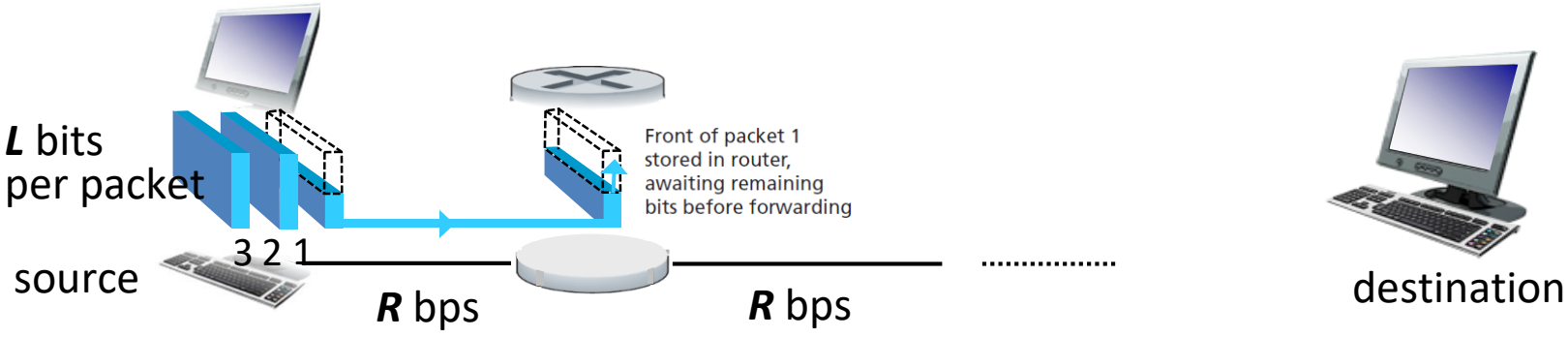
- takes *L*/*R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- *store and forward:* entire packet must arrive at router before it can be transmitted on next link
- end-end delay = 2*L*/*R* (assuming zero propagation delay)

more on delay shortly …

one-hop numerical example:

- *L* = 7.5 Mbits
- *R* = 1.5 Mbps
- one-hop transmission delay = 5 sec

Packet-switching: store-and-forward



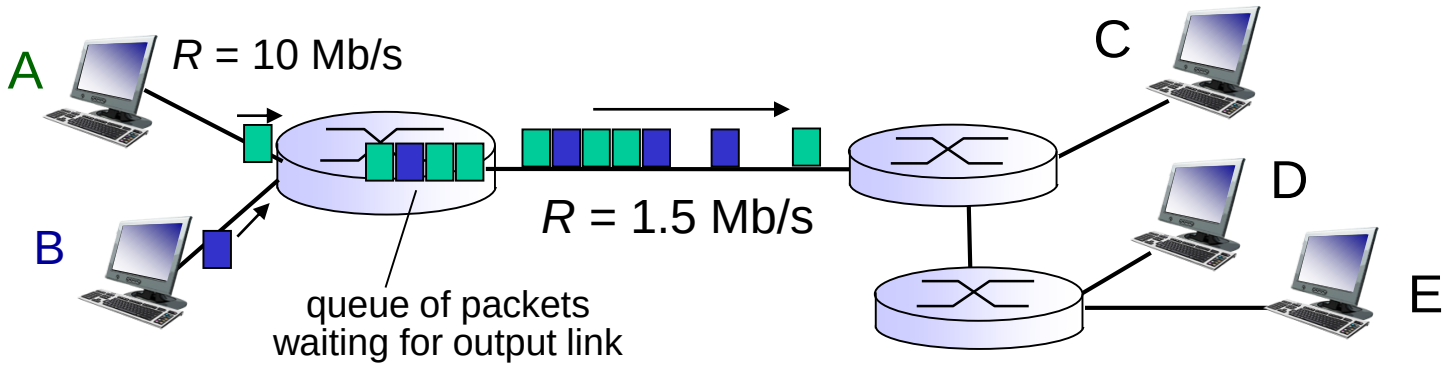
- General case of sending one packet from source to destination over a path consisting of *N* links each of rate *R*
  - *N*-1 routers between source and destination
  - End-to-end delay: $d_{\text{end-to-end}}$

$$d_{\text{end-to-end}} = N\frac{L}{R}$$

Packet Switching: queuing delay, packet loss

- Each packet switch has multiple links attached to it.
- For each attached link, the packet switch has an **output buffer** (also called an **output queue**), which stores packets that the router is about to send into that link.
  - The output buffers play a key role in packet switching.
- If an arriving packet needs to be transmitted onto a link but finds the link busy with the transmission of another packet, the arriving packet must wait in the output buffer.
  ⇒ output buffer **queuing delay**
- Since the amount of buffer space is finite, an arriving packet may find that the buffer is completely full with other packets waiting for transmission.
  ⇒ **packet loss** will occur
  - either the arriving packet or one of the already-queued packets will be dropped.

Packet Switching: queuing delay, packet loss



queuing delay & packet loss:

- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time:
  - packets will queue, wait to be transmitted on link ⇒ **queuing delay**
  - packets can be dropped (lost) if memory (buffer) fills up ⇒ **packet loss**

Forwarding Tables & Routing Protocols

- How does the router determine which link it should forward the packet onto?
  - In the Internet, every end system has an address called an **IP address**.
  - When a source end system wants to send a packet to a destination end system, the source includes the destination's IP address in the packet's header
  - When a packet arrives at a router in the network, the router examines a portion of the packet's destination address and forwards the packet to an adjacent router

Forwarding Tables & Routing Protocols

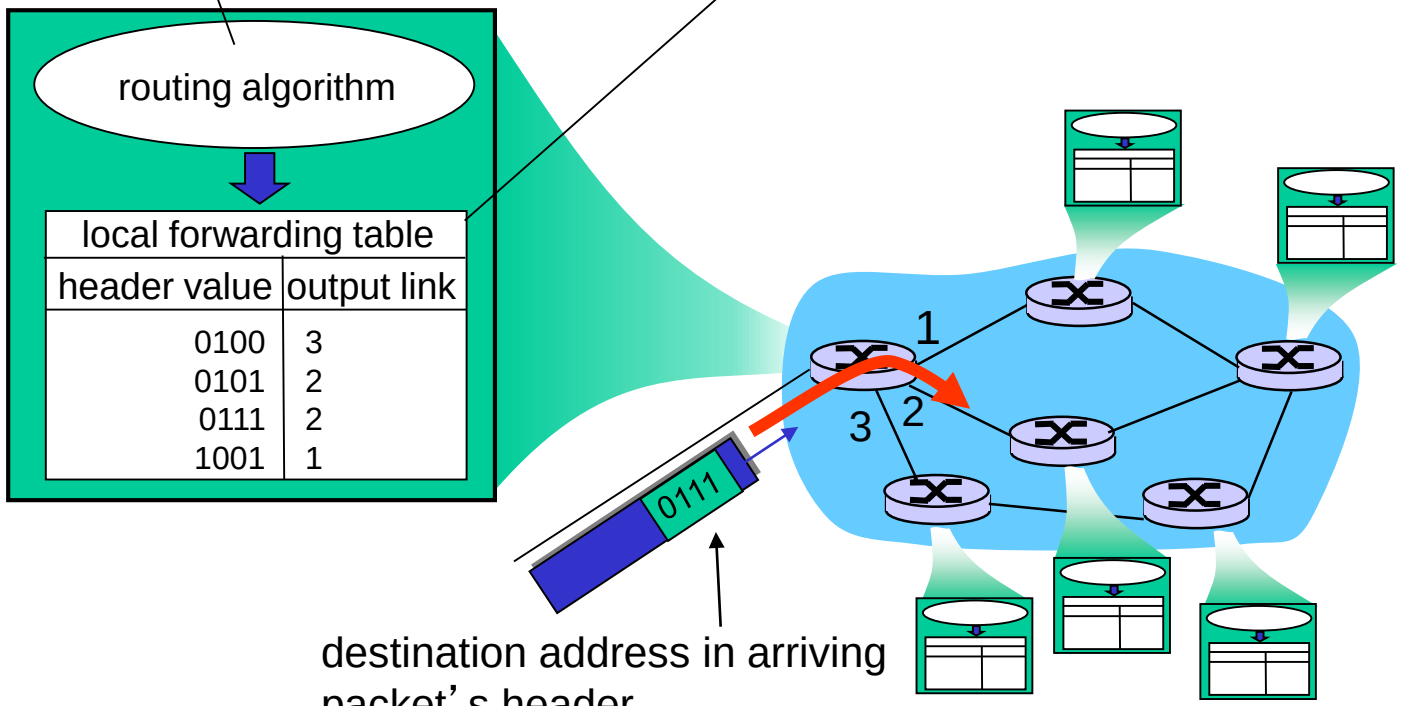
- More detail process of routing
  - each router has a **forwarding table** that maps destination addresses to that router's outbound links.
  - When a packet arrives at a router, the router examines the address and searches its forwarding table, using this destination address, to find the appropriate outbound link.
  - The router then directs the packet to this outbound link.

Forwarding Tables & Routing Protocols

routing: determines source-destination route taken by packets

- *routing algorithms (protocols)*

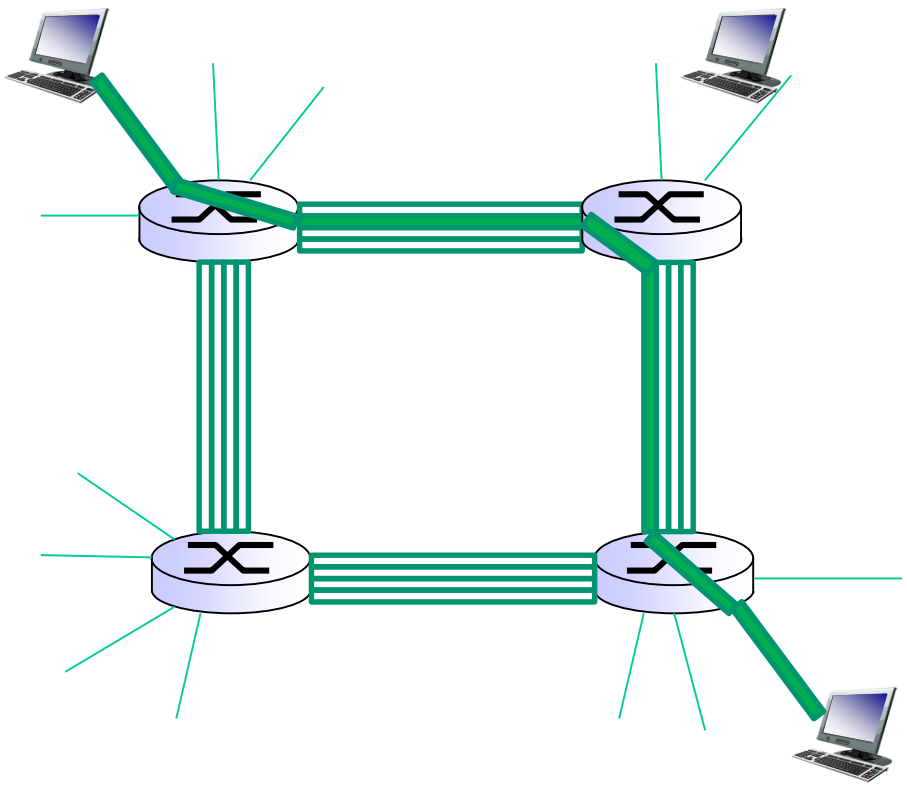
forwarding: move packets from router's input to appropriate router output



Alternative core: circuit switching

end-end resources allocated to, reserved for "call" between source & dest:

- In diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call *(no sharing)*
- Commonly used in traditional telephone networks

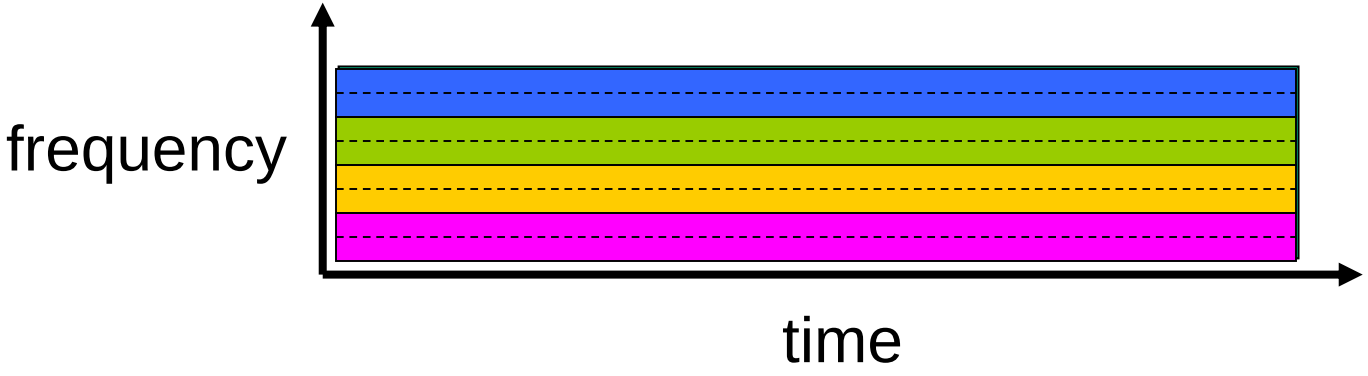


Circuit switching: FDM versus TDM

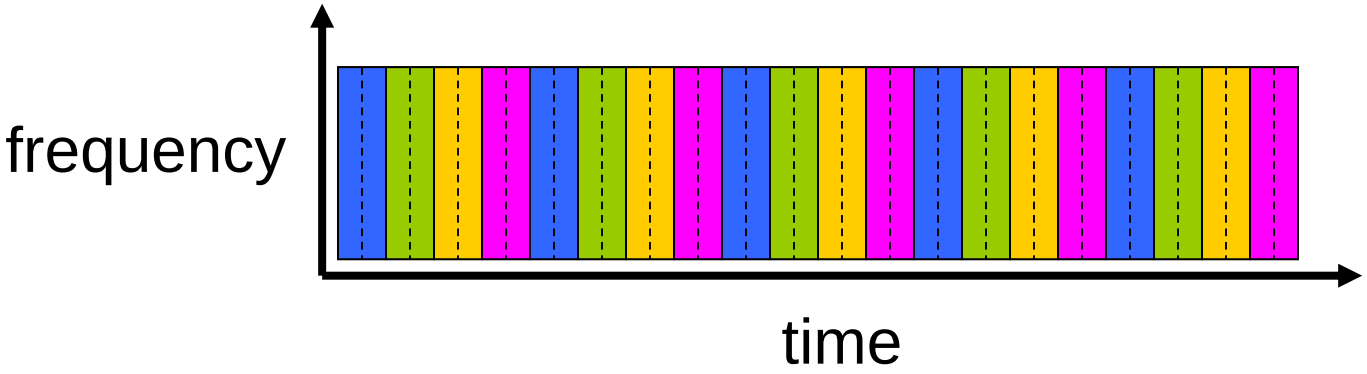
Example:

4 users

FDM



TDM

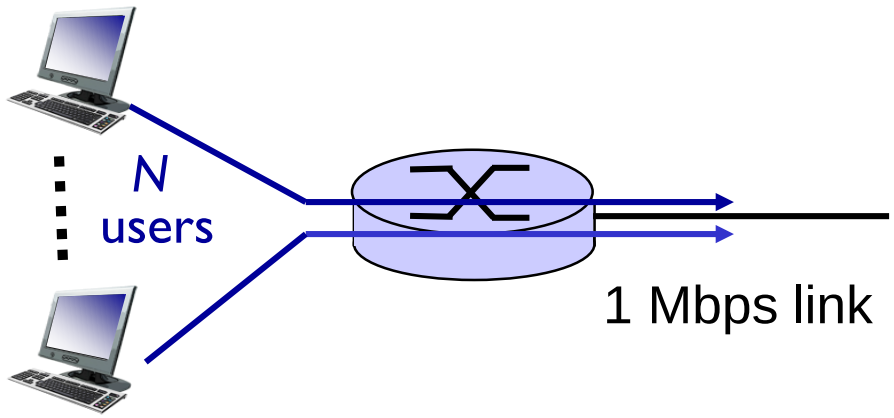


Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- 1 Mb/s link
- each user:
  - 100 kb/s when "active"
  - active 10% of time



❖ *circuit-switching:*

- 10 users

❖ *packet switching:*

- with 35 users, probability > 10 active at same time is less than 0.0004 *

- When there are 10 or fewer simultaneously active users (which happens with probability 0.9996), the aggregate arrival rate of data is less than or equal to 1 Mbps, the output rate of the link.
- Thus, when there are 10 or fewer active users, users' packets flow through the link essentially without delay, as is the case with circuit switching

* Check out the online interactive exercises for more examples

Packet switching versus circuit switching

is packet switching a "slam dunk winner?"

- great for bursty data
  - resource sharing
  - simpler, no call setup
- excessive congestion possible: packet delay and loss
  - protocols needed for reliable data transfer, congestion control
- *Q:* How to provide circuit-like behavior?
  - bandwidth guarantees needed for audio/video apps
  - still an unsolved problem (chapter 7)

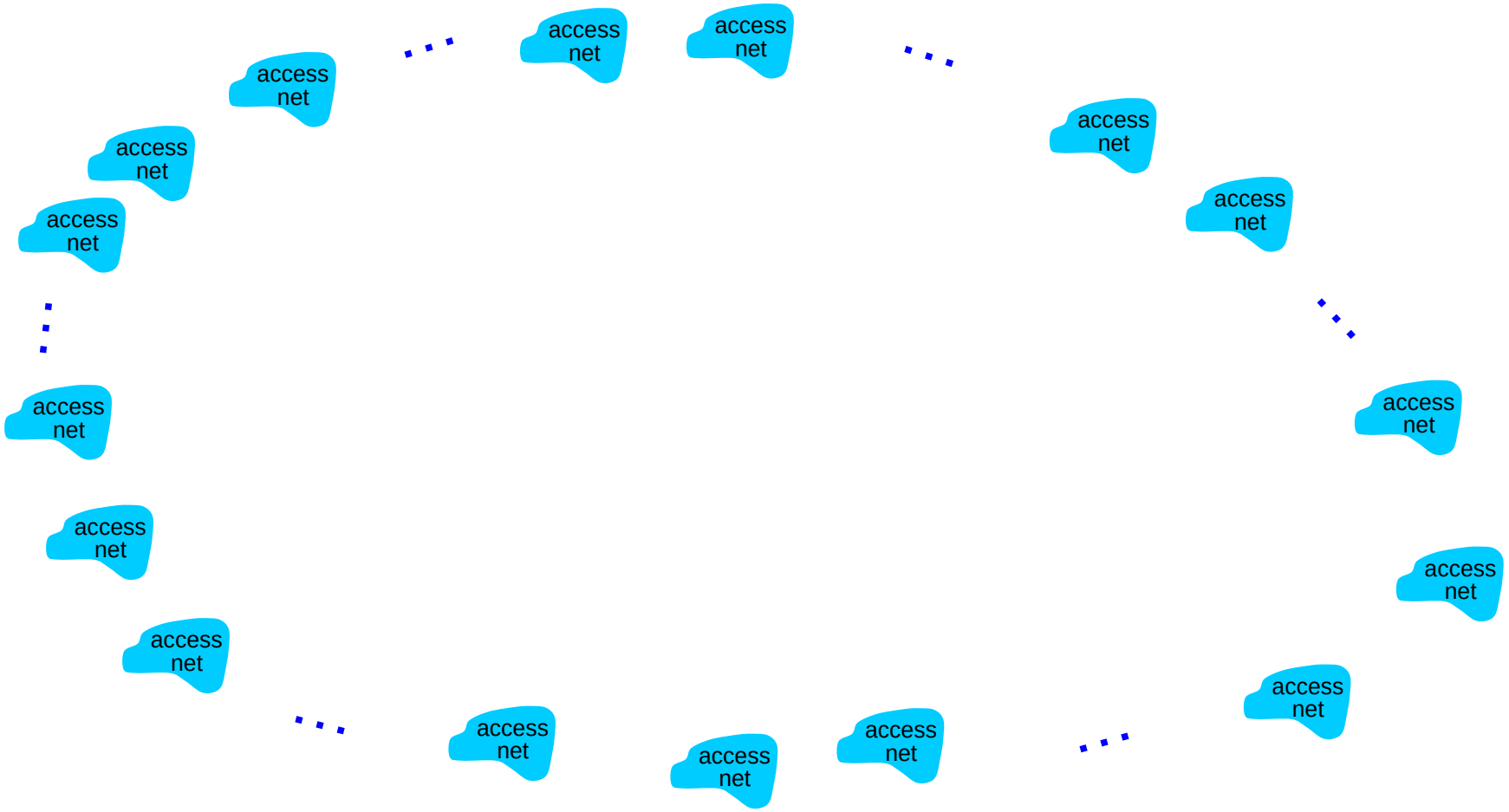
Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?

Internet structure: network of networks

- End systems connect to Internet via access ISPs (Internet Service Providers)
  - Residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - So that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - Evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

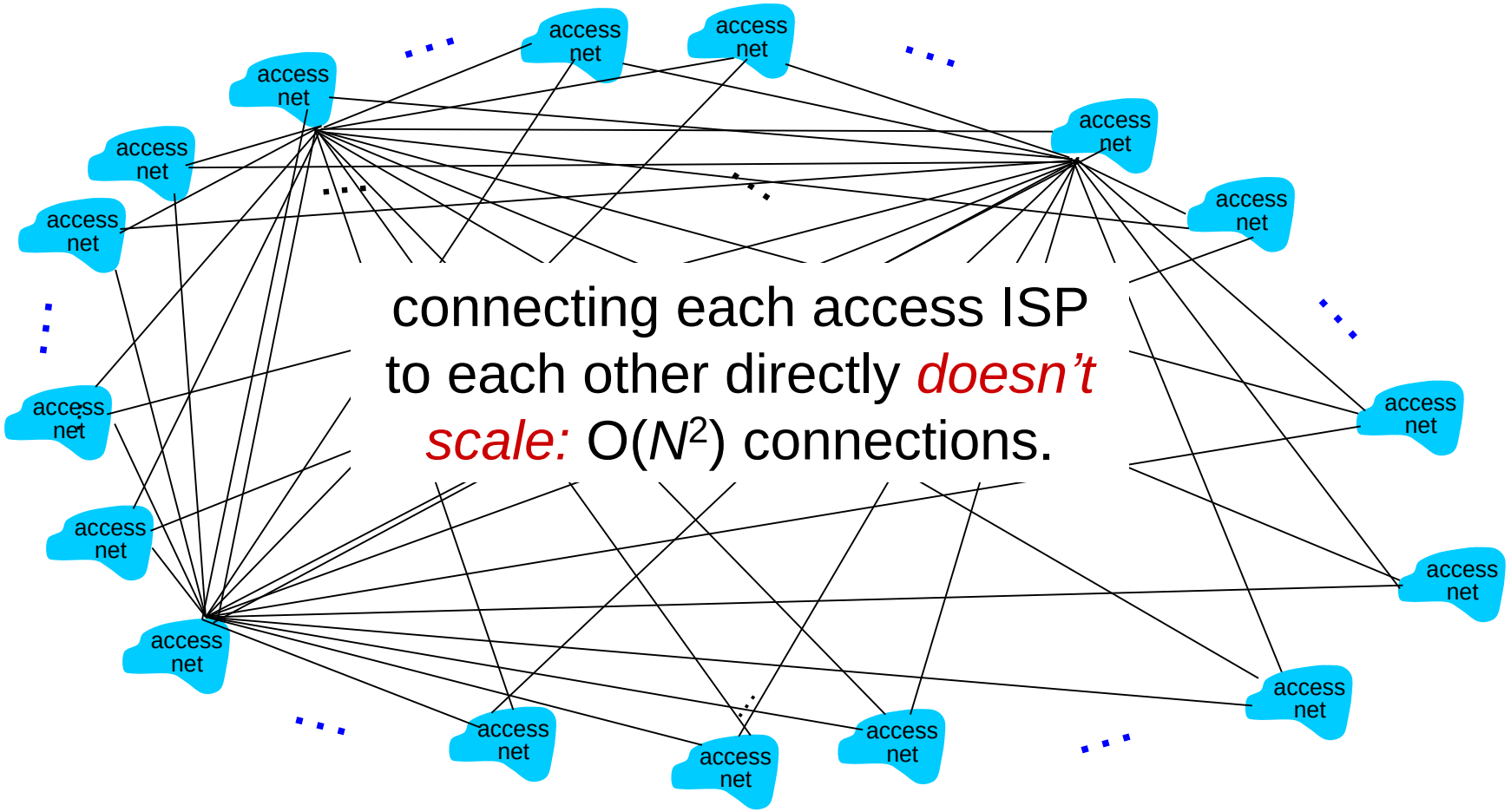
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: network of networks

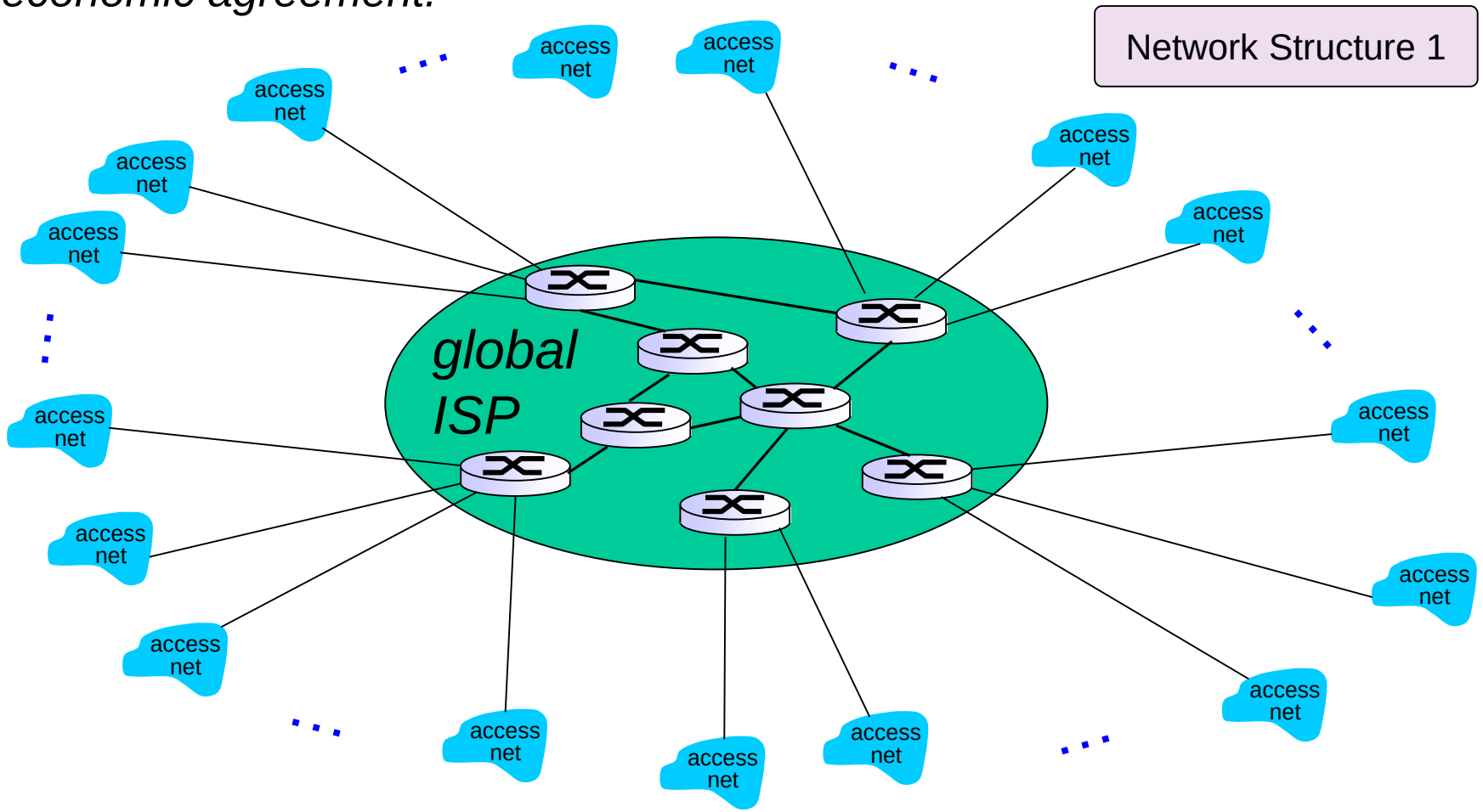
Option: connect each access ISP to every other access ISP?



Internet structure: network of networks

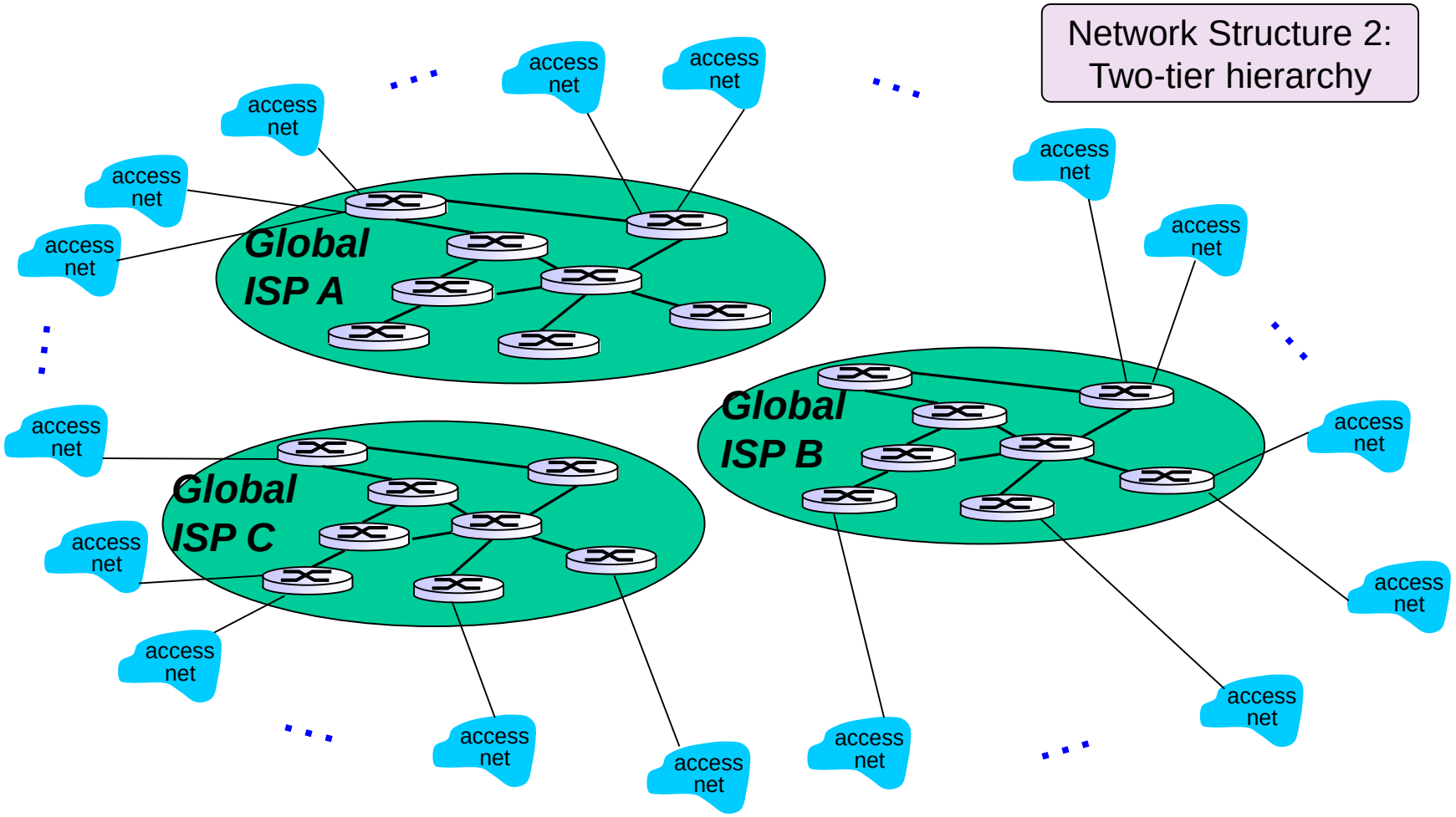
Option: connect each access ISP to a global transit ISP?

Customer ISP(access ISP) and provider ISP(global ISP) have economic agreement.



Internet structure: network of networks

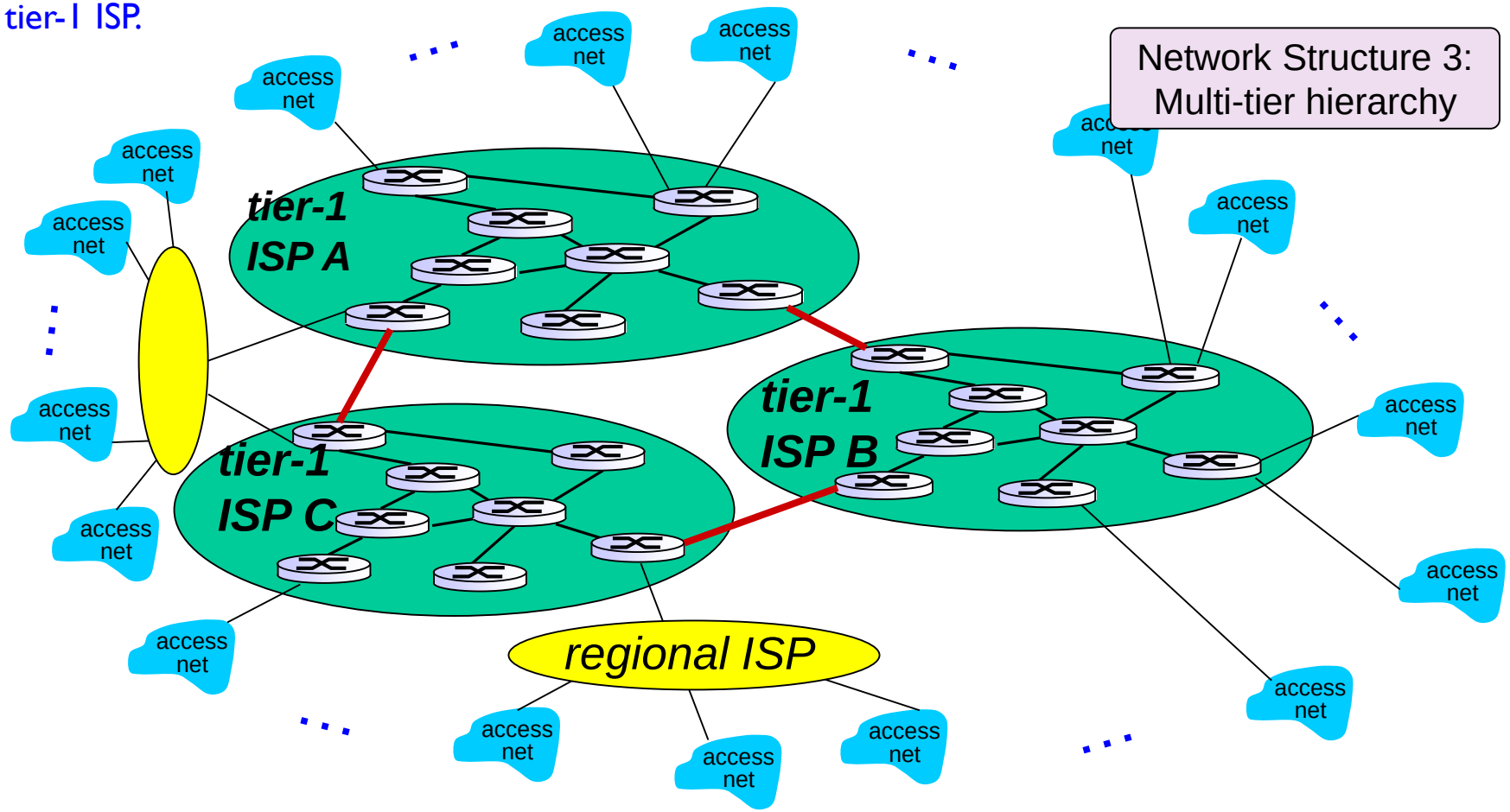
But if one global ISP is viable business, there will be competitors.
⇒ Hundreds of thousands of access ISPs and multiple global transit ISPs.



Internet structure: network of networks

Regional networks may arise to connect access nets to ISPs.

There may be a larger regional ISP (possibly spanning an entire country) to which the smaller regional ISPs in that region connect; the larger regional ISP then connects to a tier-1 ISP.



Internet structure: network of networks

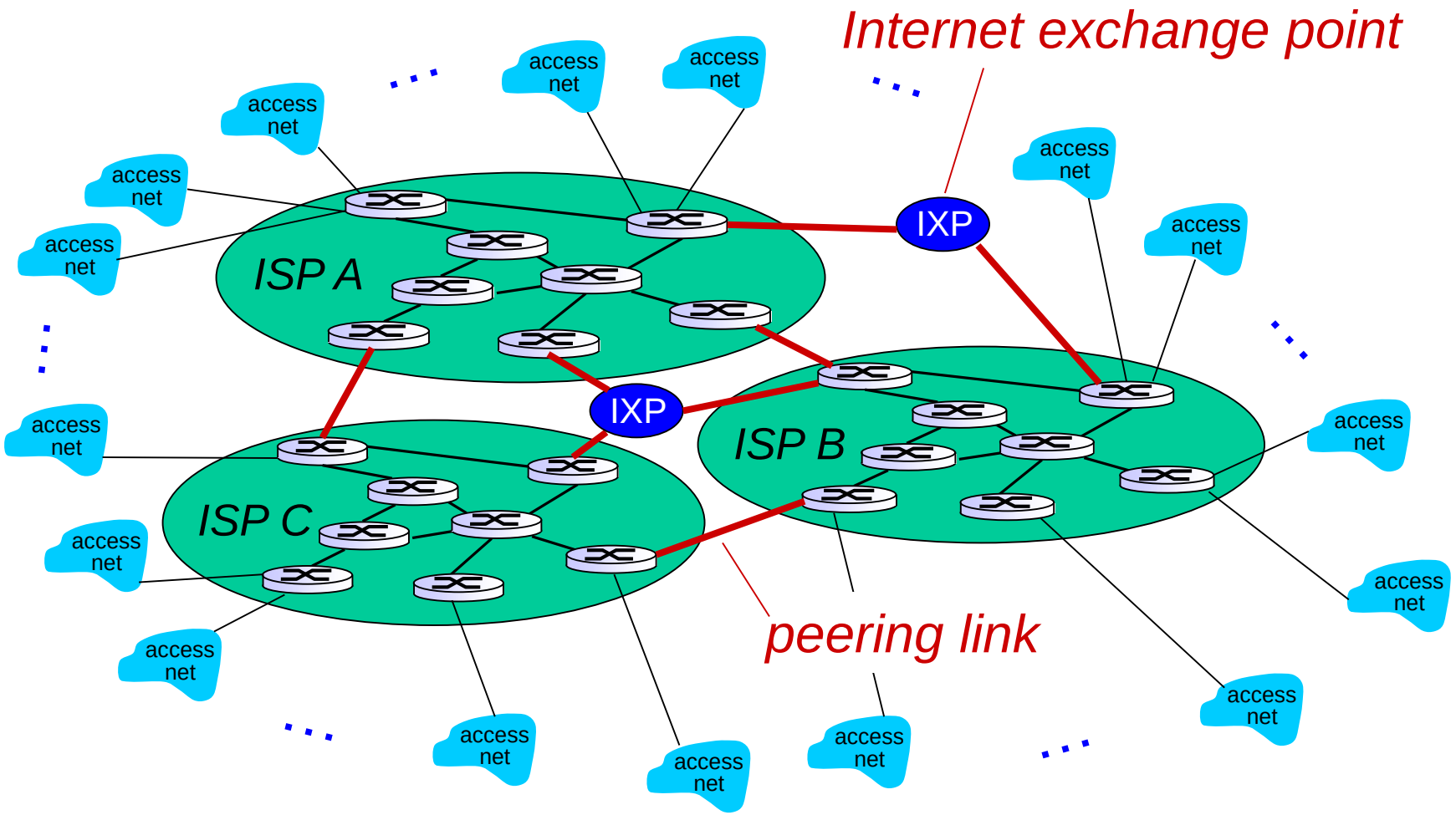
- To build a network that more closely resembles today's Internet, we must add points of presence (PoPs), multi-homing, peering, and Internet exchange points (IXPs) to the hierarchical Network Structure 3.
- Points of Presence (PoPs)
  - PoPs exist in all levels of the hierarchy, except for the bottom (access ISP) level.
    - A **PoP** is simply <u>a group of one or more routers (at the same location)</u> in the provider's network where customer ISPs can connect into the provider ISP.
- Multi-homing
  - Any ISP (except for tier-1 ISPs) may choose to **multi-home**, that is, <u>to connect to two or more provider ISPs</u>.
    - Ex) an access ISP may multi-home with two regional ISPs, or it may multi-home with two regional ISPs and also with a tier-1 ISP.
    - Similarly, a regional ISP may multi-home with multiple tier-1 ISPs.

Internet structure: network of networks

- Multi-homing (Continued)
  - When an ISP multi-homes, it can continue to send and receive packets into the Internet even if one of its providers has a failure.
- Peer
  - A pair of nearby ISPs at the same level of the hierarchy can **peer**, that is, they can directly connect their networks together so that all the traffic between them passes over the direct connection rather than through upstream intermediaries.
  - When two ISPs peer, it is typically settlement-free, that is, neither ISP pays the other.
  - Tier-1 ISPs also peer with one another, settlement-free
- IXP (Internet Exchange Point)
  - a third-party company can create an **IXP** (typically in a stand-alone building with its own switches), which is a meeting point where multiple ISPs can peer together.
  - There are roughly 300 IXPs in the Internet today.

Internet structure: network of networks

We refer to this ecosystem—consisting of access ISPs, regional ISPs, tier-1 ISPs, PoPs, multi-homing, peering, and IXPs—as **Network Structure 4**.



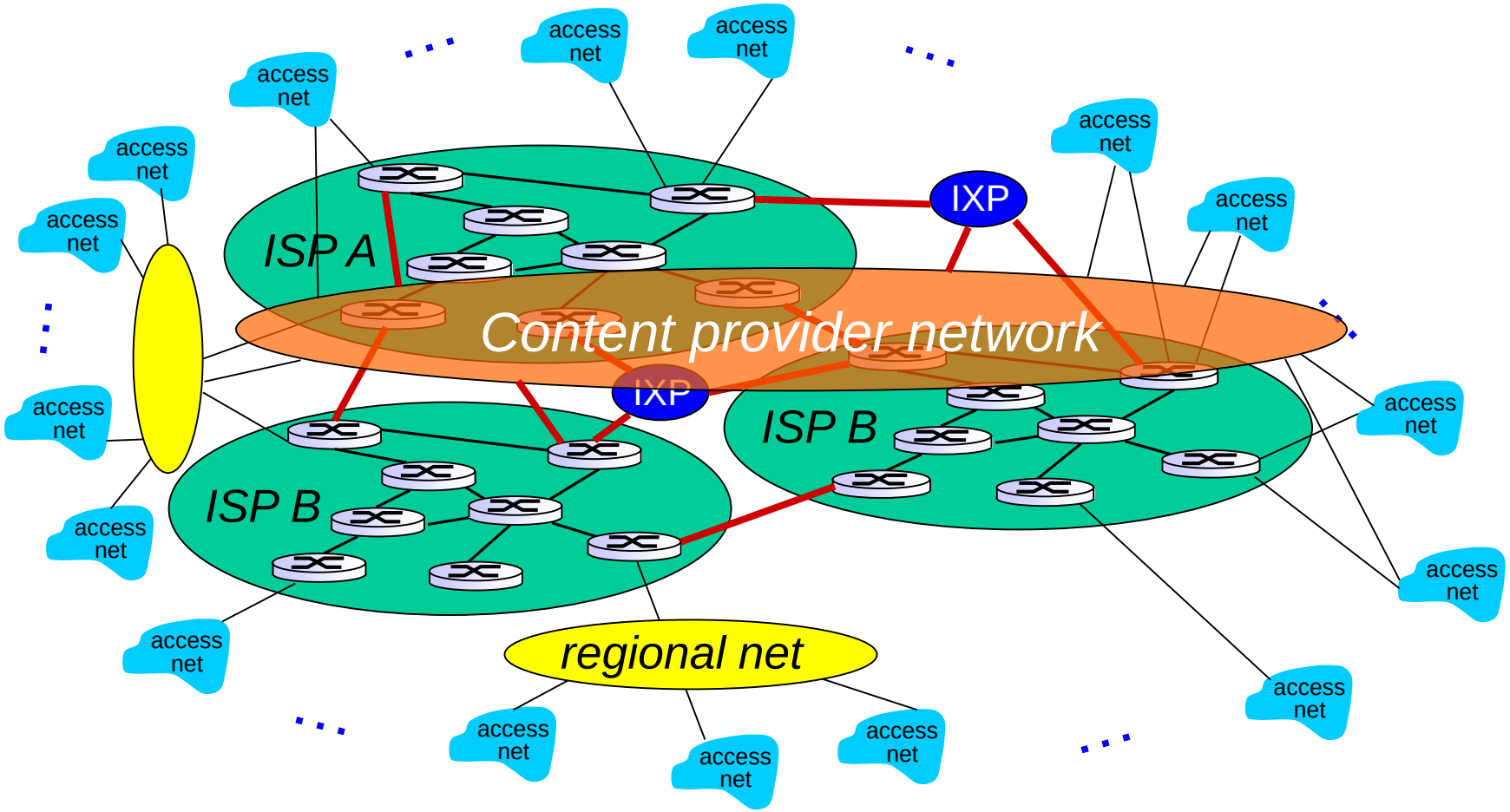
Internet structure: network of networks

- Content Provider Networks
  - Google is currently one of the leading examples of such a content provider network.
  - As of this writing, it is estimated that Google has 30 to 50 data centers distributed across North America, Europe, Asia, South America, and Australia.
  - Some of these data centers house over one hundred thousand servers, while other data centers are smaller, housing only hundreds of servers.
  - The Google data centers are all interconnected via Google's private TCP/IP network, which spans the entire globe but is nevertheless separate from the public Internet.
  - Importantly, the Google private network only carries traffic to/from Google servers
  - By creating its own network, a content provider not only reduces its payments to upper-tier ISPs, but also has greater control of how its services are ultimately delivered to end users.

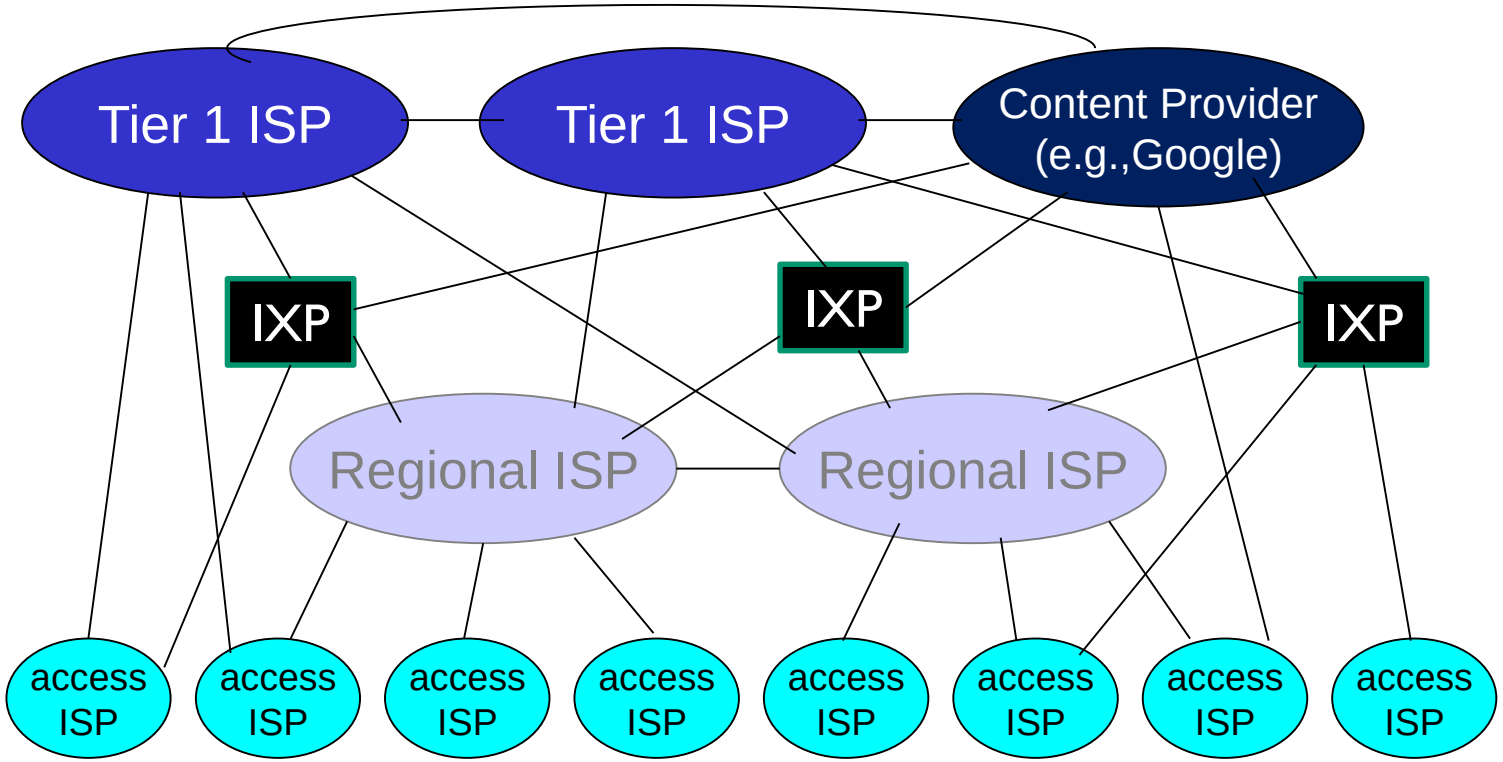
Internet structure: network of networks

… and content provider networks (e.g., Google, Microsoft) may run their own network, to bring services, content close to end users

Network Structure 4 + Content Provider Networks = **Network Structure 5**

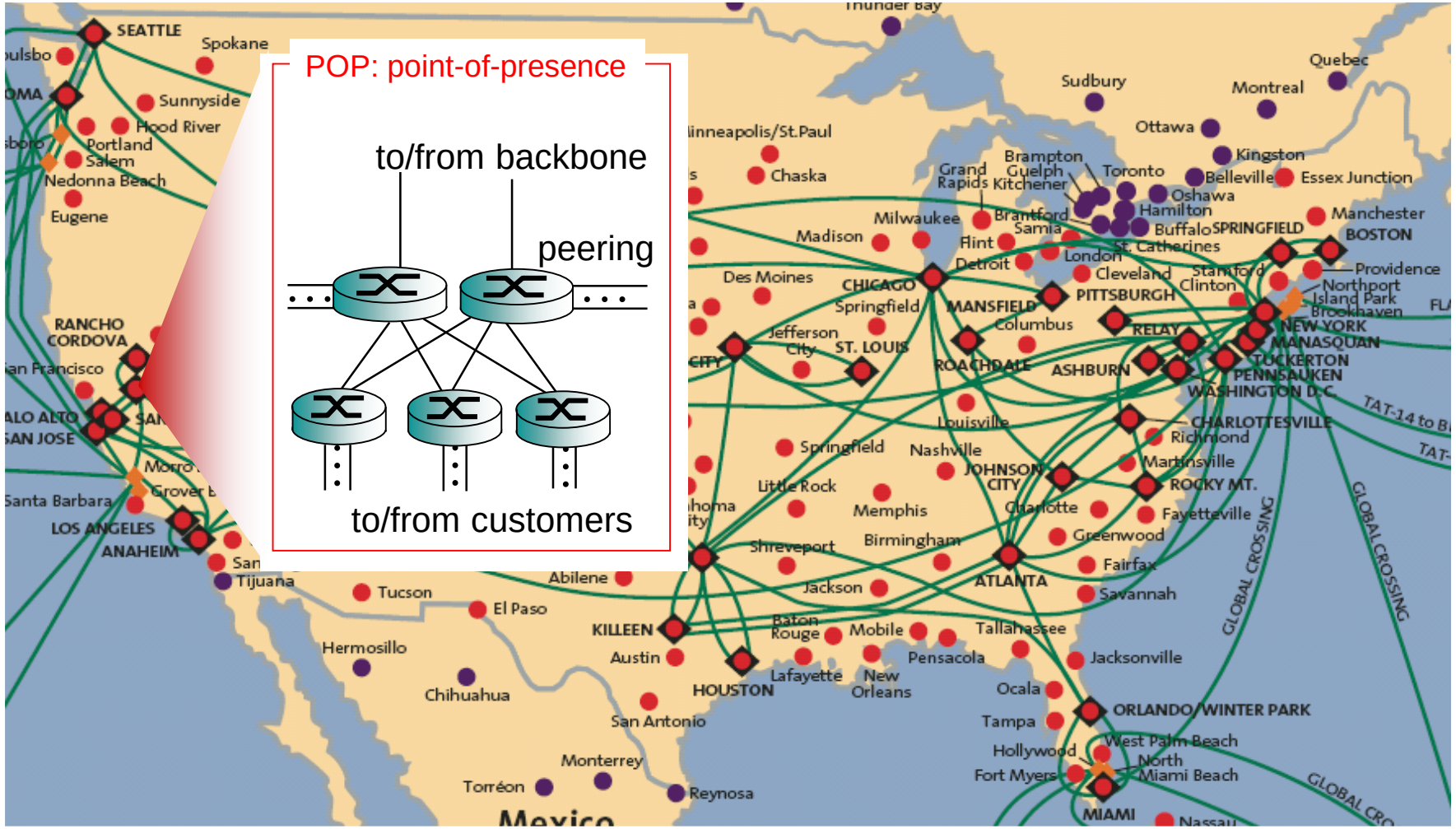


Internet structure: network of networks



- at center: small # of well-connected large networks
  - "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
  - content provider network (e.g, Google): private network that connects it data centers to Internet, often bypassing tier-1, regional ISPs

Tier-1 ISP: e.g., Sprint



Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

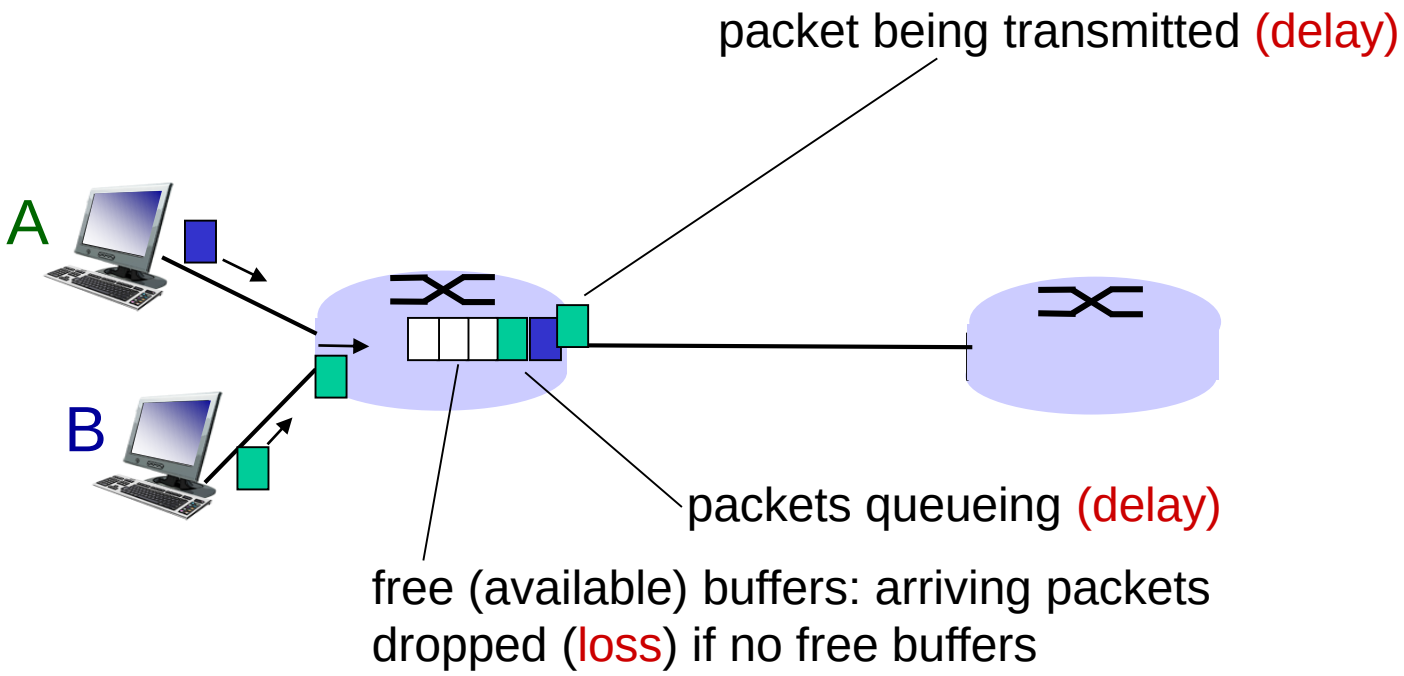
1.6 networks under attack: security

1.7 history

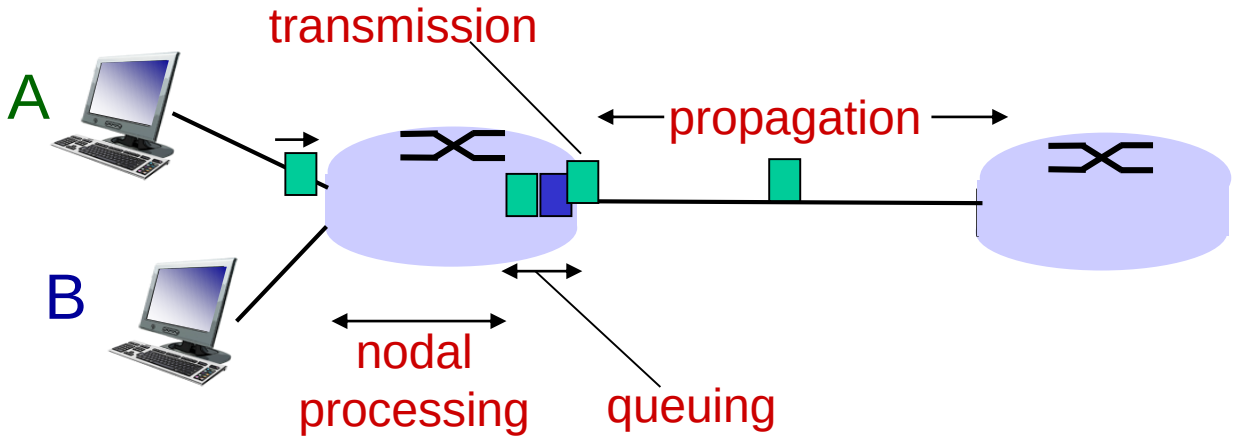
How do loss and delay occur?

packets *queue* in router buffers

- packet arrival rate to link (temporarily) exceeds output link capacity
- packets queue, wait for turn



Four sources of packet delay

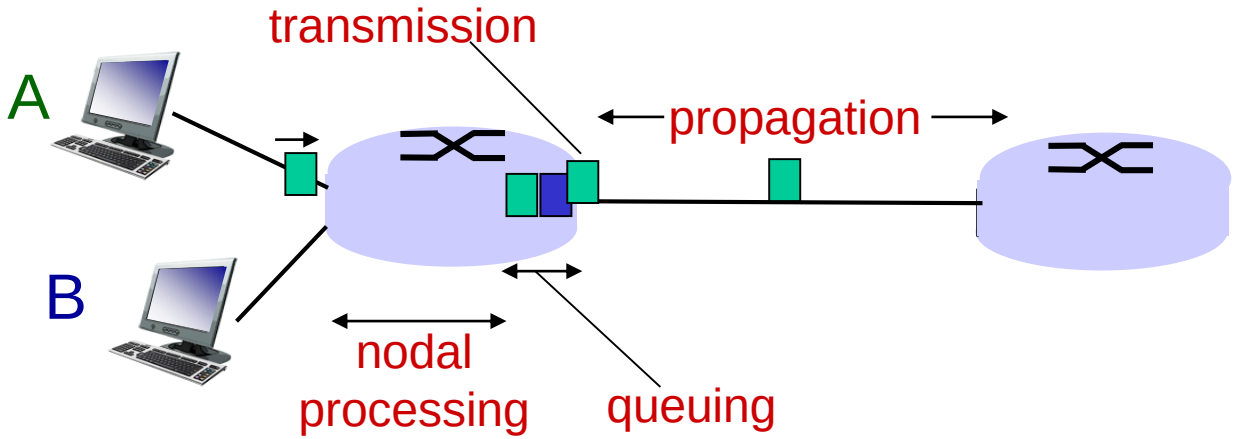


$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{proc}: nodal processing delay

- Time required to examine the packet's header and determine where to direct the packet
- Time needed to check bit-level errors
- typically on the order of <u>microseconds or less</u>

Four sources of packet delay

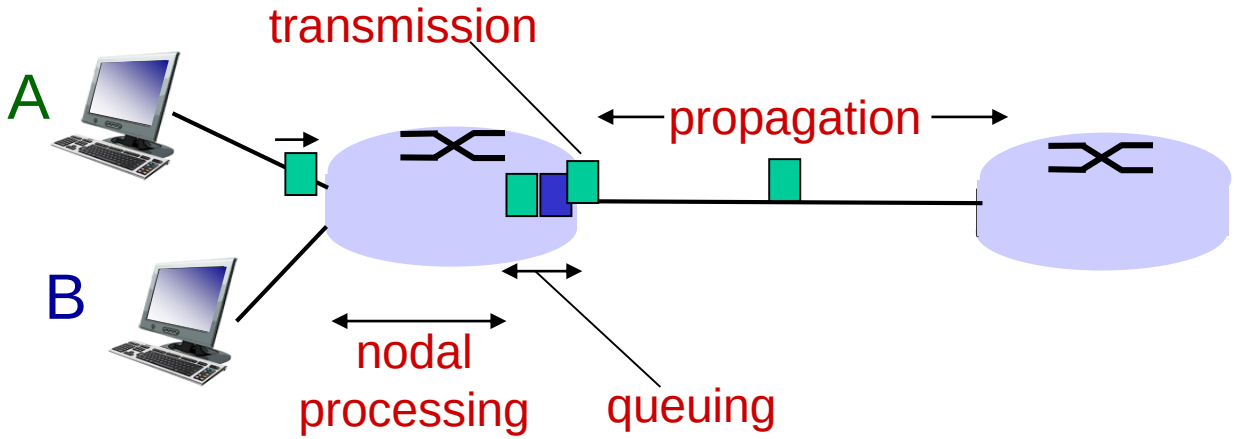


$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{queue}: queuing delay

- time waiting at output link for transmission
- depends on congestion level of router
- If the queue is empty and no other packet is currently being transmitted, then our packet's queuing delay will be zero.
- on the order of microseconds to milliseconds in practice

Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay:

- *L*: packet length (bits)
- *R*: link *bandwidth (bps)*
- $d_{trans} = L/R$
- *on the order of microseconds to milliseconds in practice*

d_{prop}: propagation delay:

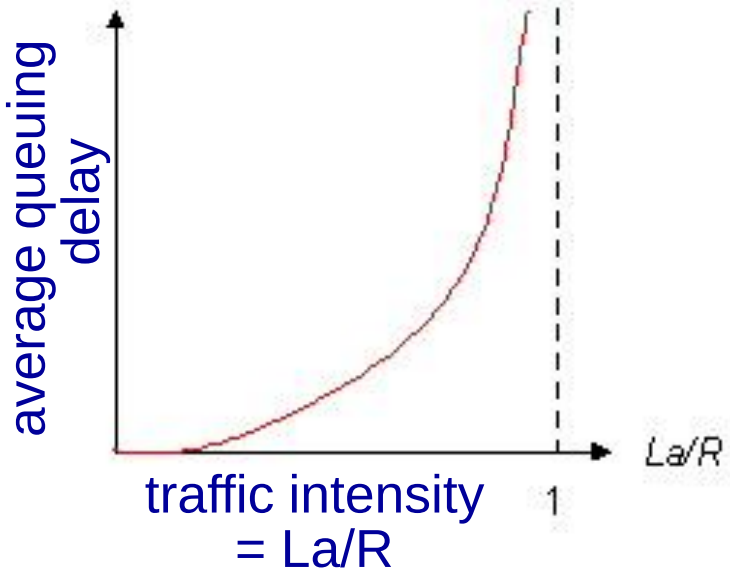
- *d*: length of physical link
- *s*: propagation speed in medium (2×10^8 m/sec ~ 3×10^8 m/sec)
- $d_{prop} = d/s$
- *on the order of milliseconds*

Queuing delay

- The most complicated and interesting component of nodal delay is the queuing delay.
- In fact, queuing delay is so important and interesting in computer networking that thousands of papers and numerous books have been written about it
- Unlike the other three delays (namely, d_{proc}, d_{trans}, and d_{prop}), the queuing delay can vary from packet to packet.
  - Ex) if 10 packets arrive at an empty queue at the same time,
    - the first packet transmitted will suffer no queuing delay,
    - the last packet transmitted will suffer a relatively large queuing delay
      (while it waits for the other nine packets to be transmitted).
- Thus, in characterizing queuing delay, one typically uses statistical measures, such as:
  average queuing delay, variance of queuing delay, and the probability that the queuing delay exceeds some specified value.

Queuing delay

- *R:* link bandwidth (bps)
- *L:* packet length (bits)
- a: average packet arrival rate
- **Traffic intensity: La/R**



- *La/R* ~ 0 : avg. queuing delay small
- *La/R* →1: avg. queuing delay large
- *La/R* > 1: more "work" arriving than can be serviced, queuing delay → infinity

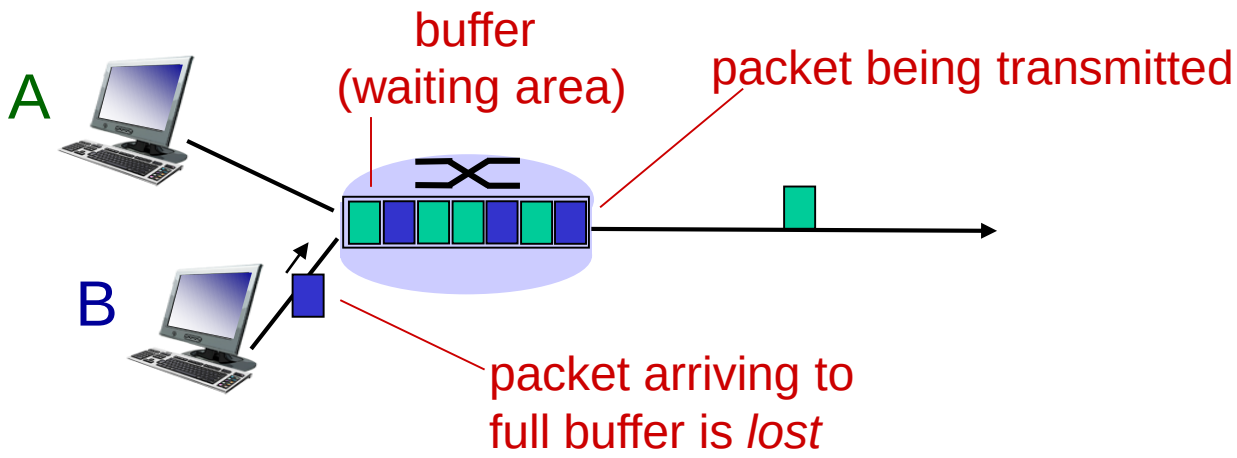
La/R ~ 0

La/R -> 1

* Check out the Java applet for an interactive animation on queuing and loss

Packet loss

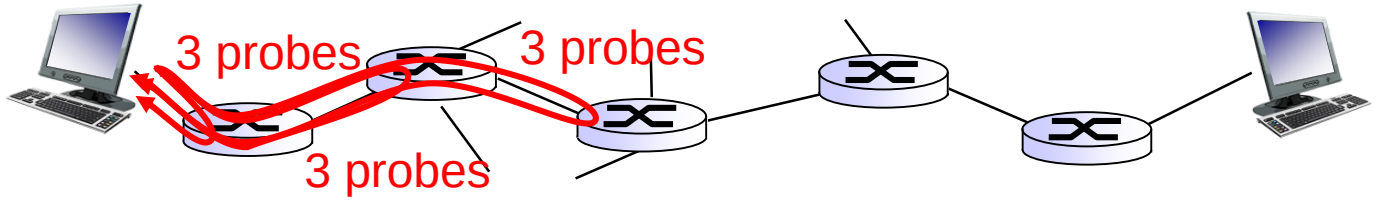
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue will be **drop**ped (aka **lost**)
- lost packet may be retransmitted by previous node, by source end system, or not at all
- The fraction of lost packets increases as the traffic intensity increases.
- Therefore, performance at a node is often measured not only in terms of delay, but also in terms of the probability of packet loss.



* Check out the Java applet for an interactive animation on queuing and loss

"Real" Internet delays and routes

- what do "real" Internet delay & loss look like?
- `traceroute` program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three packets that will reach router *i* on path towards destination
  - router *i* will return packets to sender
  - sender times interval between transmission and reply.
  - if the source receives fewer than three messages from any given router (due to packet loss in the network), Traceroute places an asterisk just after the router number and reports fewer than three round-trip times for that router.
  - Because the queuing delay is varying with time, the round-trip delay of packet n sent to a router n can sometimes be longer than the round-trip delay of packet n+1 sent to router n+1.



"Real" Internet delays, routes

traceroute: gaia.cs.umass.edu to cis.poly.edu

```
1 cs-gw (128.119.240.254) 1.009 ms 0.899 ms 0.993 ms
2 128.119.3.154 (128.119.3.154) 0.931 ms 0.441 ms 0.651 ms
3 border4-rt-gi-1-3.gw.umass.edu (128.119.2.194) 1.032 ms 0.484 ms 0.451 ms
4 acr1-ge-2-1-0.Boston.cw.net (208.172.51.129) 10.006 ms 8.150 ms 8.460 ms
5 agr4-loopback.NewYork.cw.net (206.24.194.104) 12.272 ms 14.344 ms 13.267 ms
6 acr2-loopback.NewYork.cw.net (206.24.194.62) 13.225 ms 12.292 ms 12.148 ms
7 pos10-2.core2.NewYork1.Level3.net (209.244.160.133) 12.218 ms 11.823 ms 11.793 ms
8 gige9-1-52.hsipaccess1.NewYork1.Level3.net (64.159.17.39) 13.081 ms 11.556 ms 13.297 ms
9 p0-0.polyu.bbnplanet.net (4.25.109.122) 12.716 ms 13.052 ms 12.786 ms
10 cis.poly.edu (128.238.32.126) 14.080 ms 13.035 ms 12.802 ms
```

* Do some traceroutes from exotic countries at www.traceroute.org

"Real" Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

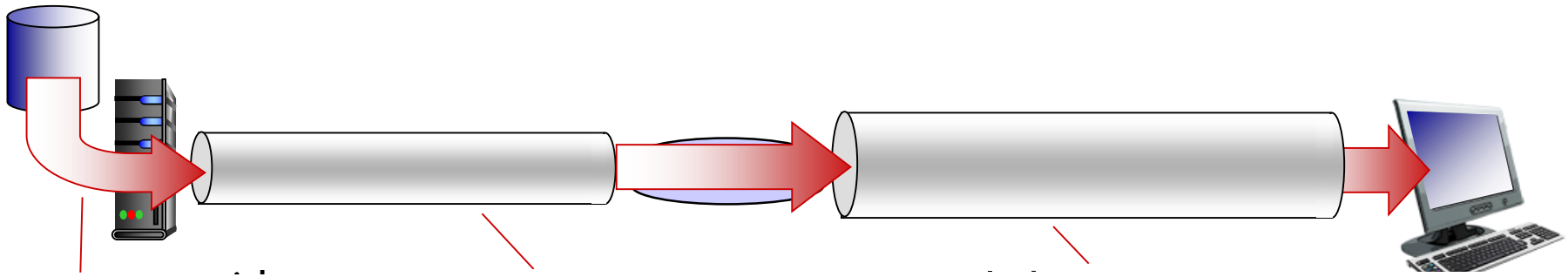
trans-oceanic link

* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

Throughput

- *throughput:* rate (bits/time unit) at which bits transferred between sender and receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time



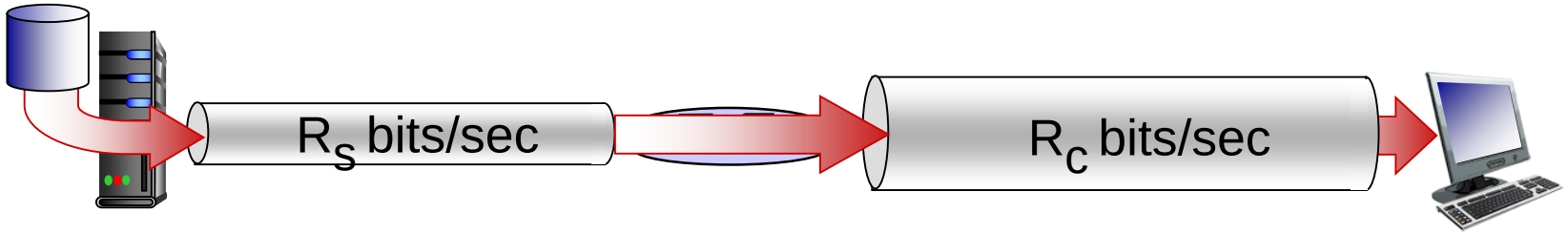
server sends bits (fluid) into pipe

pipe that can carry fluid at rate R_s bits/sec)

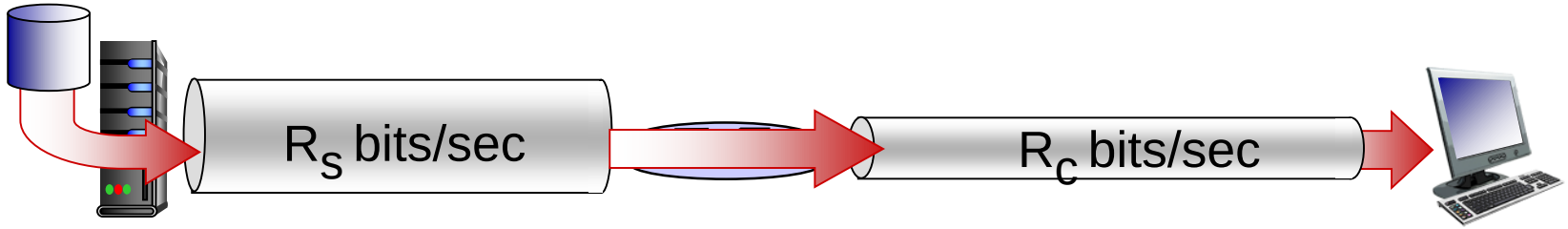
pipe that can carry fluid at rate R_c bits/sec)

Throughput (more)

- $R_s < R_c$ What is average end-end throughput?



- $R_s > R_c$ What is average end-end throughput?

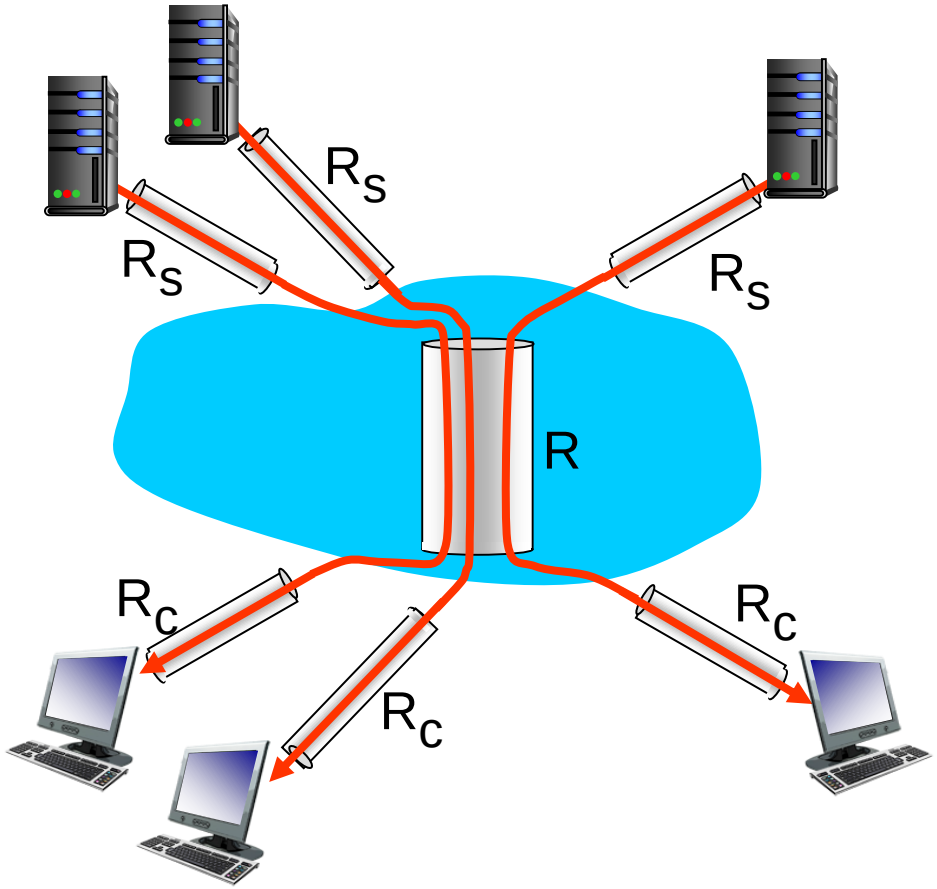


bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- per-connection end-end throughput: $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck



10 connections (fairly) share backbone bottleneck link R bits/sec

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1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Protocol "layers"

Networks are complex, with many "pieces":

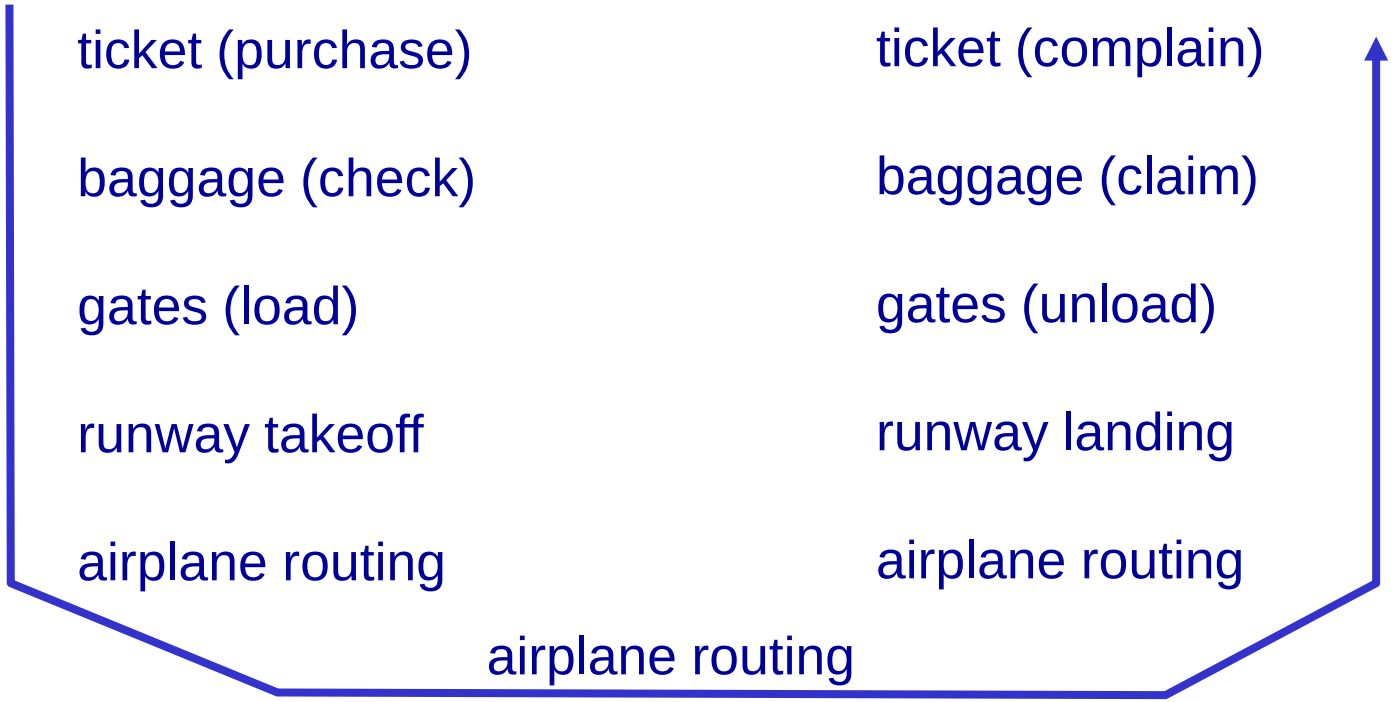
- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

is there any hope of *organizing* structure of network?

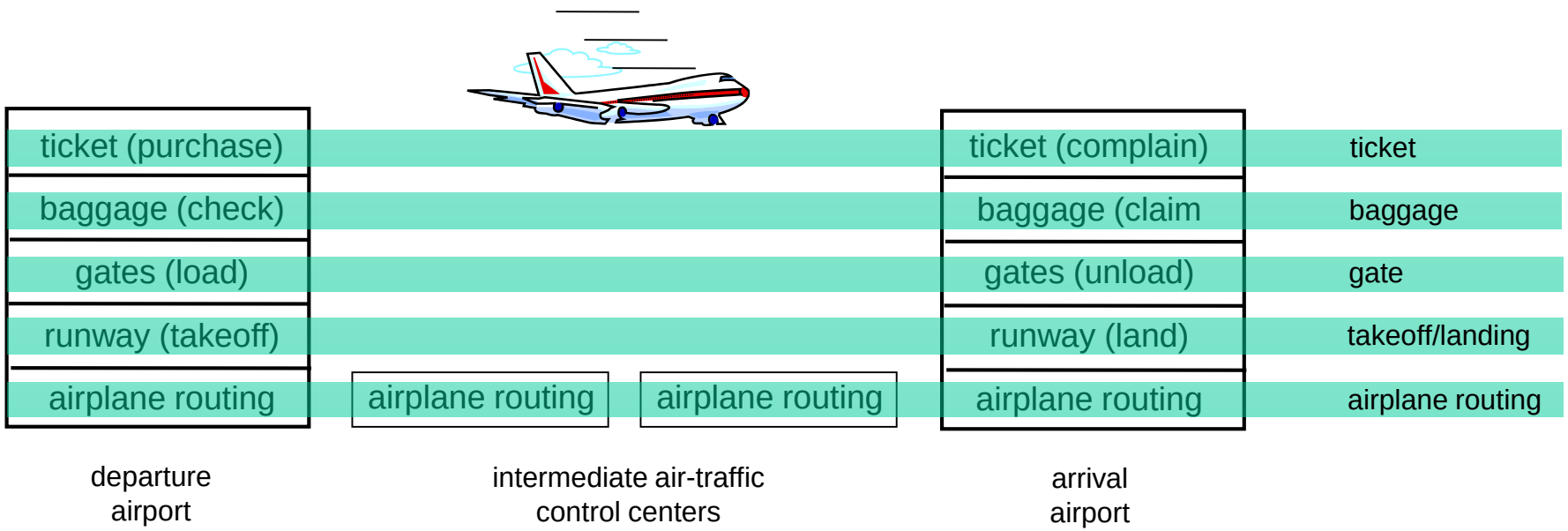
…. or at least our discussion of networks?

Organization of air travel



- a series of steps

Layering of airline functionality

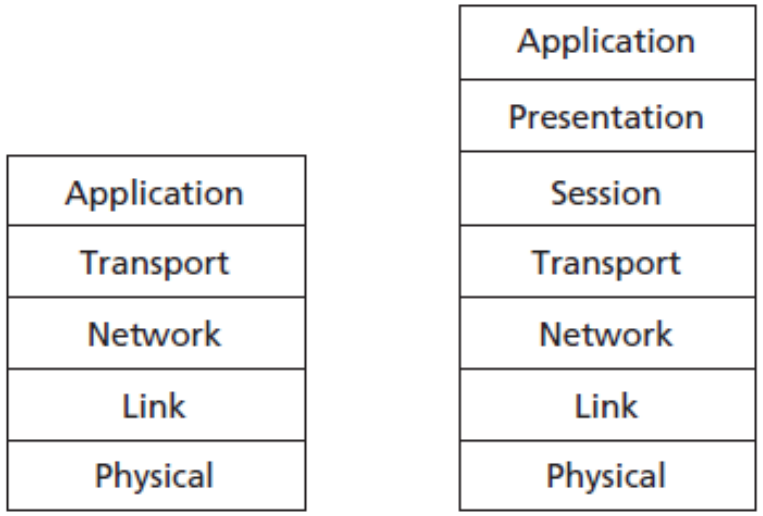


layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

Protocol Layering

- Protocol Stack
  - the protocols of the various layers
- Implementation of protocol layers
  - Software, hardware, or combination of SW and HW



a. Five-layer Internet protocol stack

b. Seven-layer ISO OSI reference model

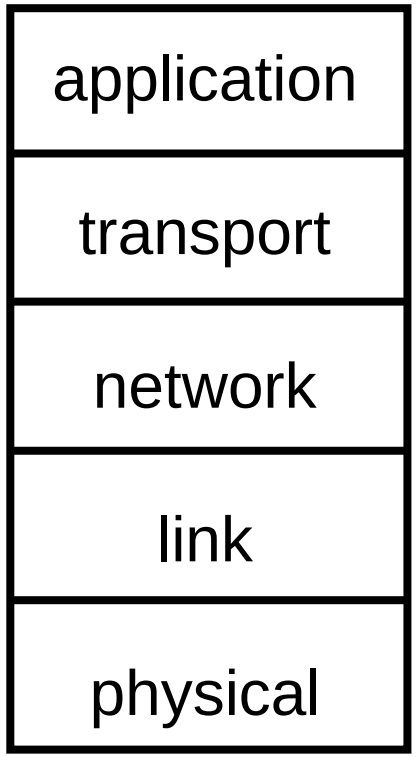
Why layering?

dealing with complex systems:

- explicit structure allows identification, relationship of complex system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change of implementation of layer's service transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system
- potential drawbacks of layering
  - one layer may duplicate lower-layer functionality.
    Ex) error recovery on a per-link basis & an end-to-end basis
  - functionality at one layer may need information that is present only in another layer. ex) a timestamp value

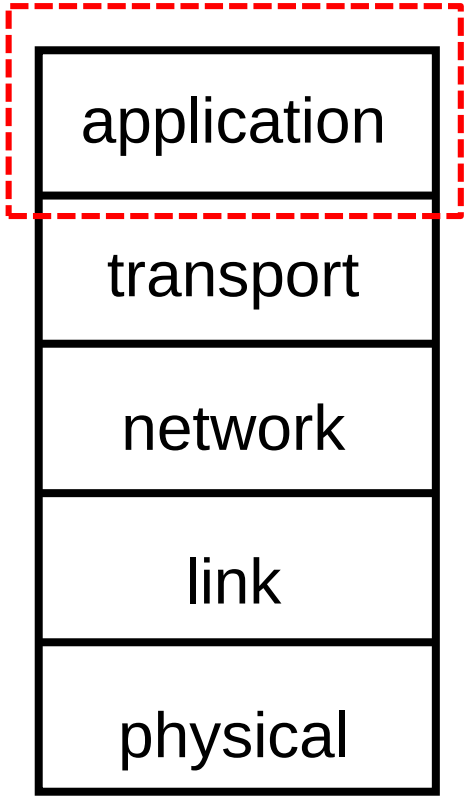
Internet protocol stack

- ***application:*** supporting network applications
  - FTP, SMTP, HTTP
- ***transport:*** transports application-layer messages between application endpoints
  - TCP, UDP
- ***network:*** routing of datagrams from source to destination
  - IP, routing protocols
- ***link:*** data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- ***physical:*** move individual bits from one node to the nest
  - Twisted-pair copper wire, coaxial cable, fiber, radio, etc.



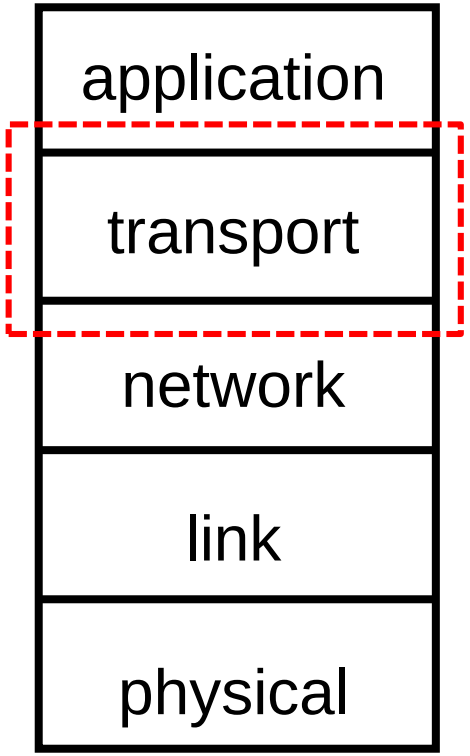
Application Layer

- The applications layer is where network applications and their application-layer protocols reside.
- Applications layer protocols
  - HTTP (Hypertext Transfer Protocol)
    - Web document request and transfer
  - SMTP (Simple Mail Transfer Protocol)
    - Transfer of e-mail messages
  - FTP (File Transfer Protocol)
    - Transfer of files between two end systems
- Packet of information at the application layer
  ⇒ **message**
  - An application-layer protocol is distributed over multiple end systems, with the application in one end system using the protocol to exchange packets of information with the application in another end system.



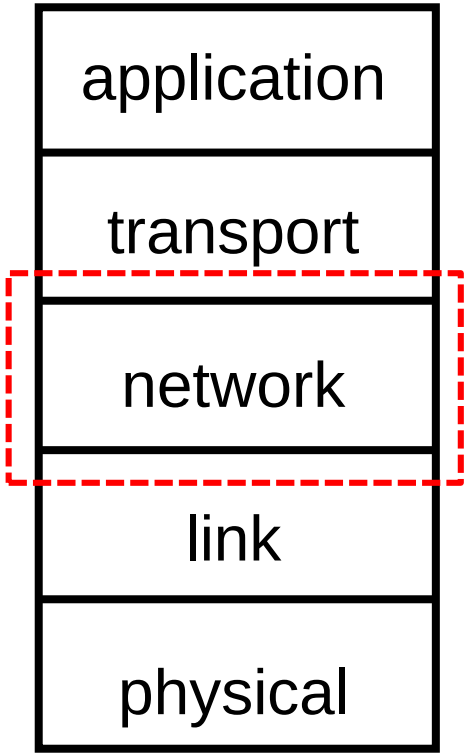
Transport Layer

- The Internet's transport layer transports application-layer messages between application endpoints.
- Transport layer protocols
  - TCP (Transmission Control Protocol)
    - a connection-oriented service to its applications
    - this service includes guaranteed delivery of application-layer messages to the destination and flow control (sender/receiver speed matching).
    - breaks long messages into shorter segments and provides a congestion-control mechanism
      ⇒ a source throttles its transmission rate when the network is congested
  - UDP (User Datagram Protocol)
    - a connectionless service to its applications
    - no reliability, no flow control, and no congestion control
- A transport-layer packet⇒ **segment**



Network Layer (1/2)

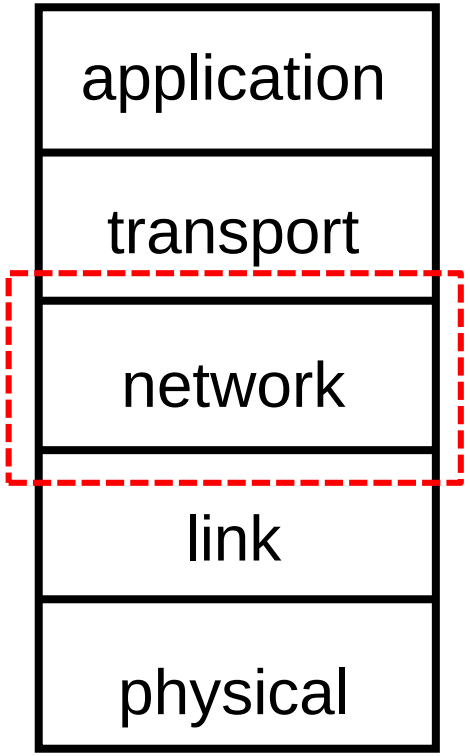
- The Internet's network layer is responsible for moving network-layer packets known as **datagrams** from one host to another.
  - TCP (or UDP) in a source host passes a transport-layer segment and a destination address to the network layer, just as you would give the postal service a letter with a destination address.
  - The network layer then provides the service of delivering the segment to the transport layer in the destination host.
- Network layer protocol ⇒ **IP Protocol**
  - defines the fields in the datagram as well as how the end systems and routers act on these fields.
  - There is only one IP protocol, and all Internet components that have a network layer must run the IP protocol.



Network Layer (2/2)

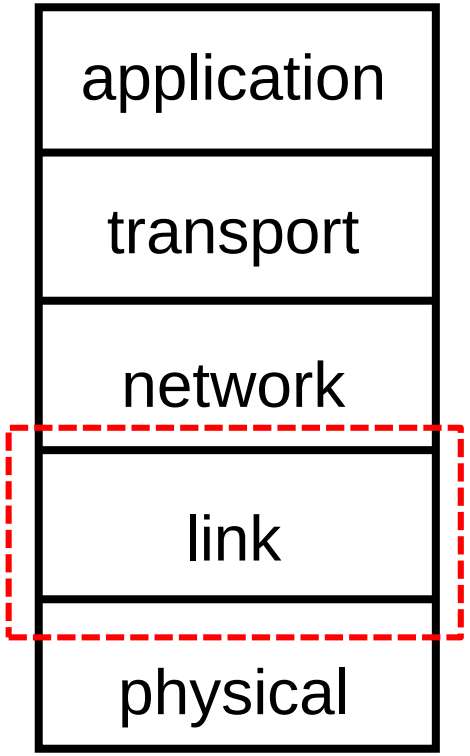
- The Internet's network layer also contains **routing protocols** that determine the routes that datagrams take between sources and destinations.
  - The Internet has many routing protocols.
  - The Internet is a network of networks
    ⇒ within a network, the network administrator can run any routing protocol desired.

- The network layer is often simply referred to as the **IP layer**, reflecting the fact that IP is the glue that binds the Internet together.



Link Layer (1/2)

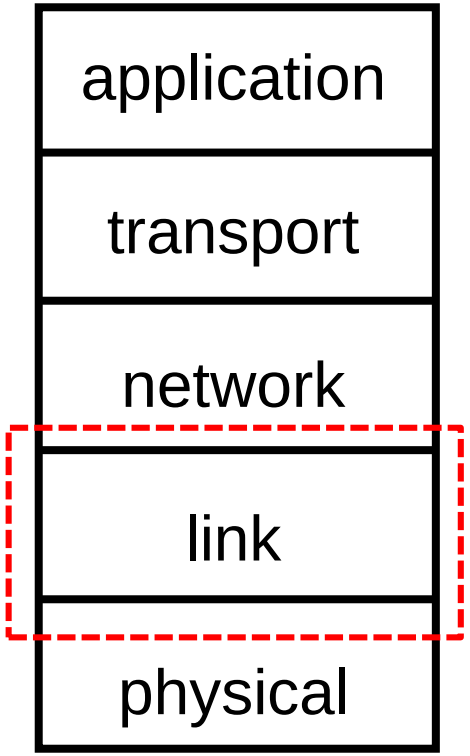
- The link layer delivers the datagram from one node to the next node along the route.
  - To move a packet from one node (host or router) to the next node in the route, the network layer relies on the services of the link layer.
- The services provided by the link layer depend on the specific link-layer protocol that is employed over the link.
  - Some link-layer protocols provide reliable delivery, from transmitting node, over one link, to receiving node.
    - this reliable delivery service is different from the reliable delivery service of TCP, which provides reliable delivery from one end system to another.
- Examples of link layer protocols
  - include Ethernet, WiFi, and the cable access network's DOCSIS protocol.



Link Layer (2/2)

- As datagrams typically need to traverse several links to travel from source to destination, a datagram may be handled by different link-layer protocols at different links along its route.
  - Ex) a datagram may be handled by Ethernet on one link and by PPP on the next link.
  - The network layer will receive a different service from each of the different link-layer protocols.

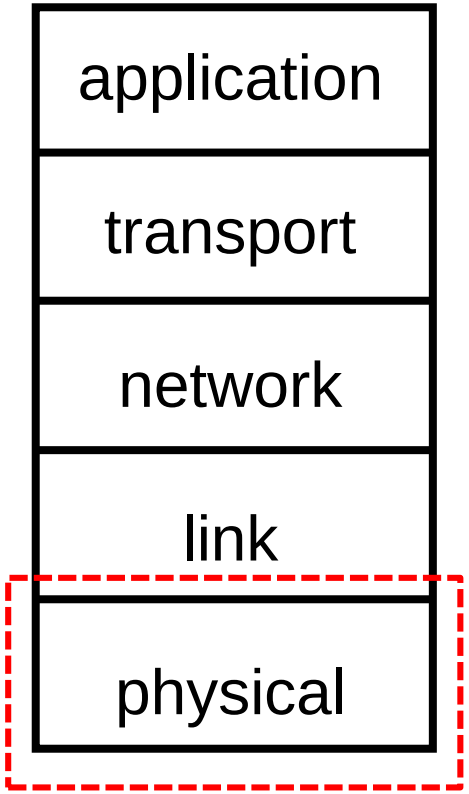
- In the textbook, the link-layer packets are called: ⇒ **frames**.



Physical Layer

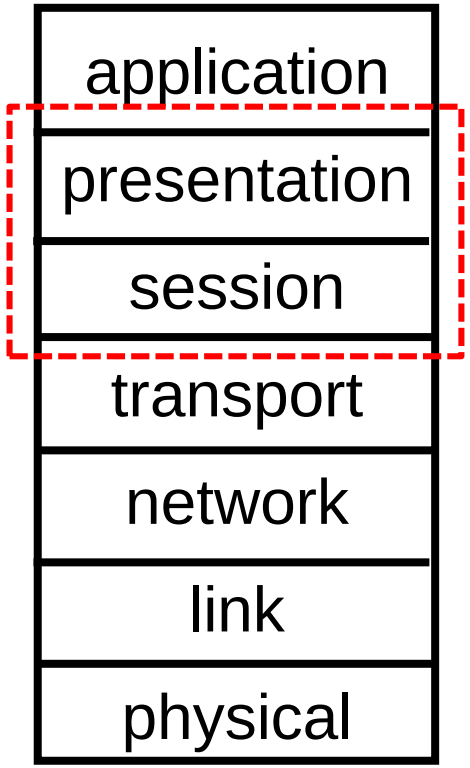
- The physical layer moves the individual bits within the frame from one node to the next.
  - the job of the link layer is to move entire frames from one network element to an adjacent network element

- The protocols in this layer are again link dependent and further depend on the actual transmission medium of the link such as twisted-pair copper wire, single-mode fiber optics.
  - Ex) Ethernet has many physical-layer protocols:
    one for twisted-pair copper wire,
    another for coaxial cable,
    another for fiber, and so on.
  - In each case, a bit is moved across the link in a different way.



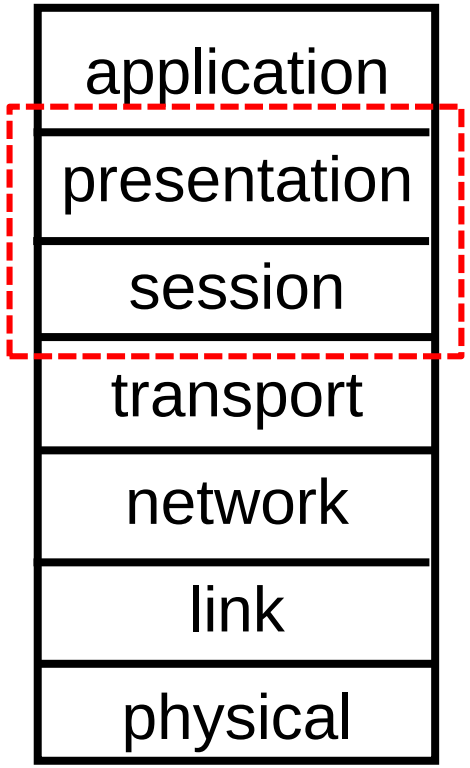
ISO/OSI reference model

- **Open Systems Interconnection (OSI) model**
  - Proposed in the late 1970's by the International Organization for Standardization (ISO)
  - Computer networks be organized around seven layers
  - Many training and university courses picked up on the ISO mandate and organized courses around the seven-layer model
  - Continues to linger on in some networking textbooks and training courses because of its early impact on networking education.
- The functionality of five of these layers is roughly the same as their similarly named Internet counterparts.
- Two additional layers present in the OSI reference model: the **presentation** layer and the **session** layer.



ISO/OSI reference model

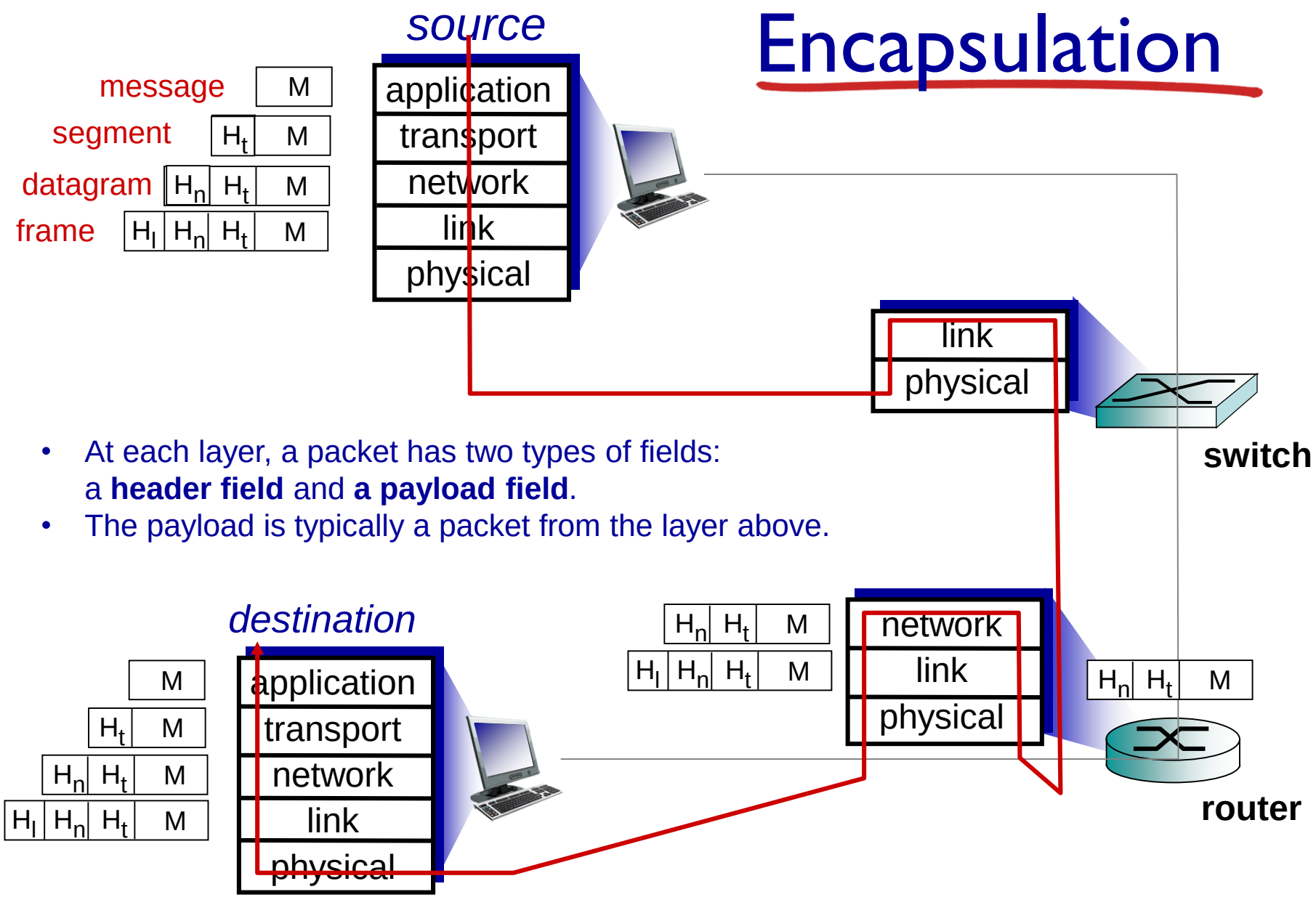
- *Presentation layer*
  - provide services that allow communicating applications to interpret the meaning of data exchanged
  - Ex) encryption, compression, machine-specific conventions
- *Session layer*
  - provides for delimiting and synchronization of data exchange, including the means to build a checkpointing and recovery scheme
- Internet stack "missing" these layers!
  - these services, *if needed*, must be implemented in application
    ⇒ it's up to the application developer.



Packet switches

- Packet Switches: Routers & Link-layer Switches
  - routers and link-layer switches organize their networking hardware and software into layers.
  - But <u>routers and link-layer switches do not implement all of the layers in the protocol stack</u>; they typically implement only the bottom layers.
- Routers vs. Link-layer switches
  - <u>link-layer switches: implement layers 1 and 2</u>
    - are capable of recognizing layer 2 addresses, such as Ethernet addresses
  - <u>routers: implement layers 1 through 3</u>
    - are capable of implementing the IP protocol (a layer 3 protocol), while link-layer switches are not.
- Note that hosts implement all five layers

Encapsulation



- At each layer, a packet has two types of fields: a **header field** and **a payload field**.
- The payload is typically a packet from the layer above.

Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Network security

- field of network security:
  - how bad guys can attack computer networks
  - how we can defend networks against attacks
  - how to design architectures that are immune to attacks

- Internet not originally designed with (much) security in mind
  - *original vision:* "a group of mutually trusting users attached to a transparent network" ☺
  - Internet protocol designers playing "catch-up"
  - security considerations in all layers!

Bad guys: put malware into hosts via Internet

- Malware can spread in the form of a virus or a worm.
  - ***virus:*** self-replicating infection
    - requires some form of user interaction to infect the user's device (e.g., e-mail attachment)
  - ***worm:*** self-replicating infection
    - can enter a device without any explicit user interaction
    - a user may be running a vulnerable network application to which an attacker can send malware
    - The worm in the newly infected device then scans the Internet, searching for other hosts running the same vulnerable network application.
    - When it finds other vulnerable hosts, it sends a copy of itself to those hosts.
- Infected hosts can be enrolled in **botnet**, used for spam, DDoS attacks.

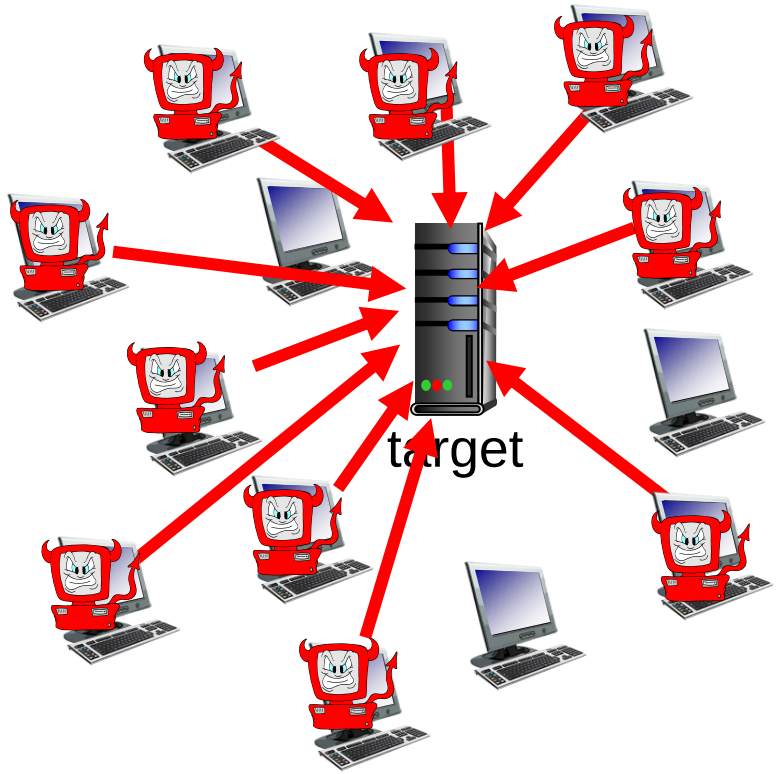
Bad guys: attack server, network infrastructure

- ***Denial of Service (DoS):***
  attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic
- Types of DoS attacks
  - Vulnerability attack
    - This involves sending a few well-crafted messages to a vulnerable application or operating system running on a targeted host.
    - If the right sequence of packets is sent to a vulnerable application or operating system, the service can stop or, worse, the host can crash.
  - Bandwidth flooding
    - The attacker sends a deluge of packets to the targeted host.
    - legitimate packets can not reach the server
  - Connection flooding
    - The attacker establishes a large number of half-open or fully open TCP connections at the target host.
    - The host can become so bogged down with these bogus connections that it stops accepting legitimate connections.

Bad guys: attack server, network infrastructure

- *Distributed Denial of Service (DDoS):*
  - the attacker controls multiple sources and has each source blast traffic at the target
  - leveraging botnets with thousands of comprised hosts

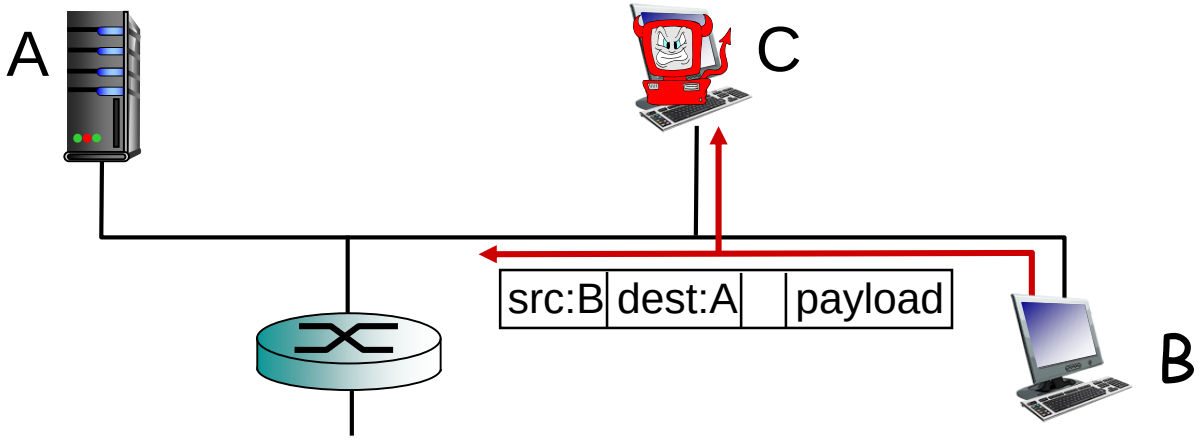
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Bad guys can sniff packets

packet "sniffing":

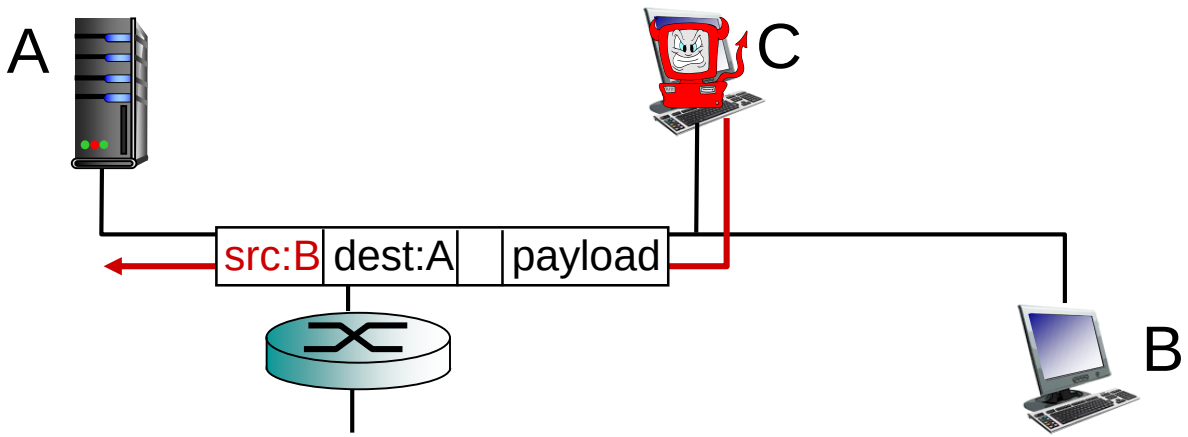
- broadcast media (shared ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



- wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: send packet with false source address



… lots more on security (throughout, Chapter 8)

Introduction: summary

covered a "ton" of material!

- Internet overview
- what's a protocol?
- network edge, core, access network
  - packet-switching versus circuit-switching
  - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security

you now have:

- context, overview, "feel" of networking
- more depth, detail *to follow!*

Thank You !